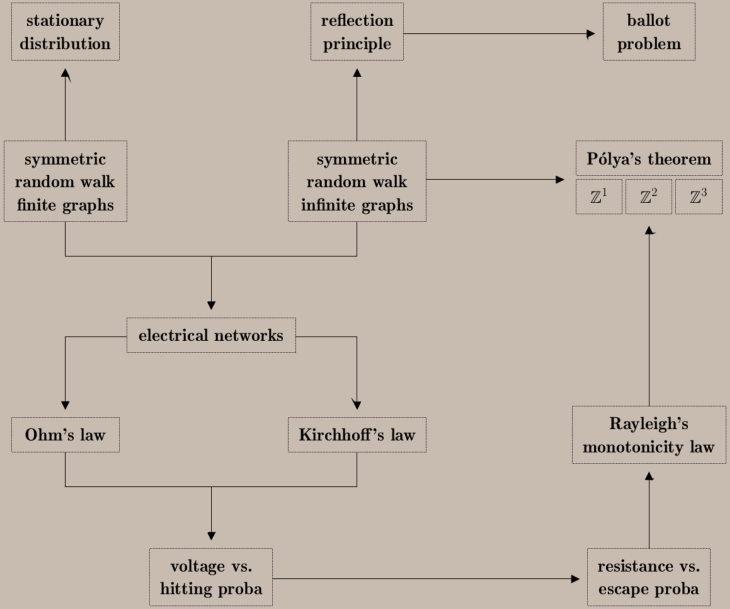
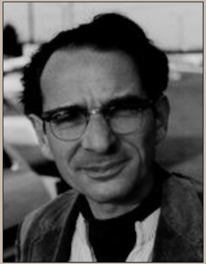
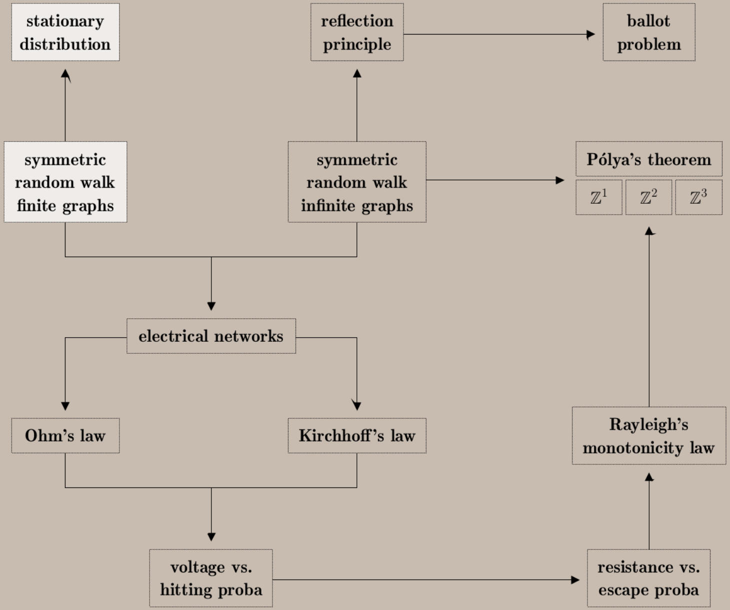



George Pólya


Frank Spitzer

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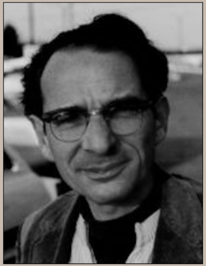
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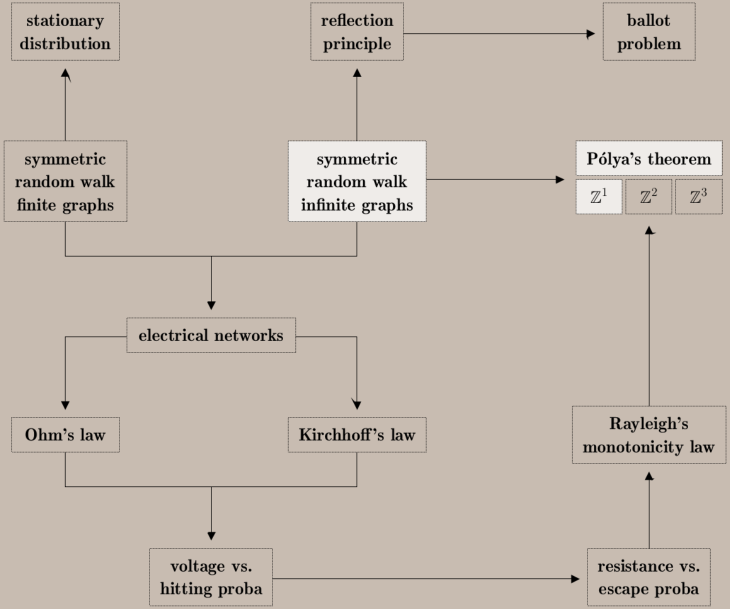

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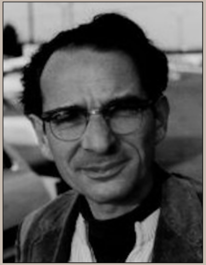

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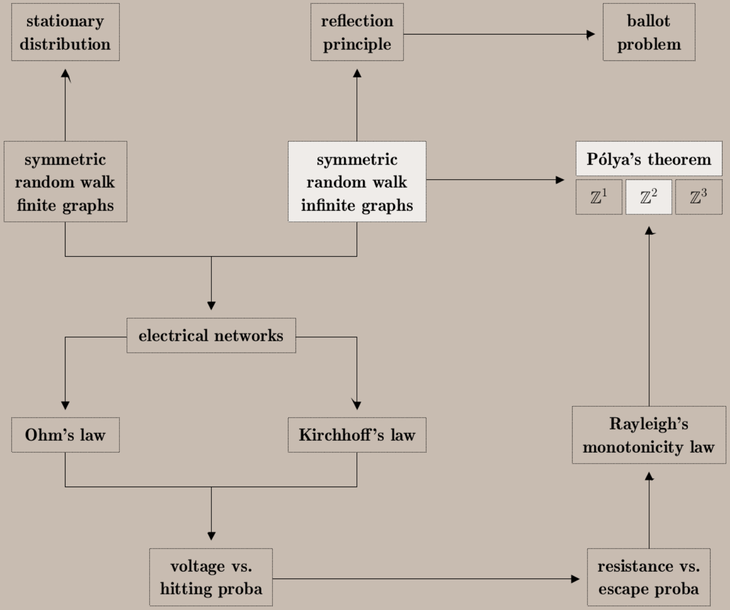

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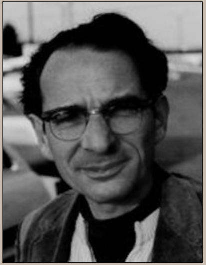

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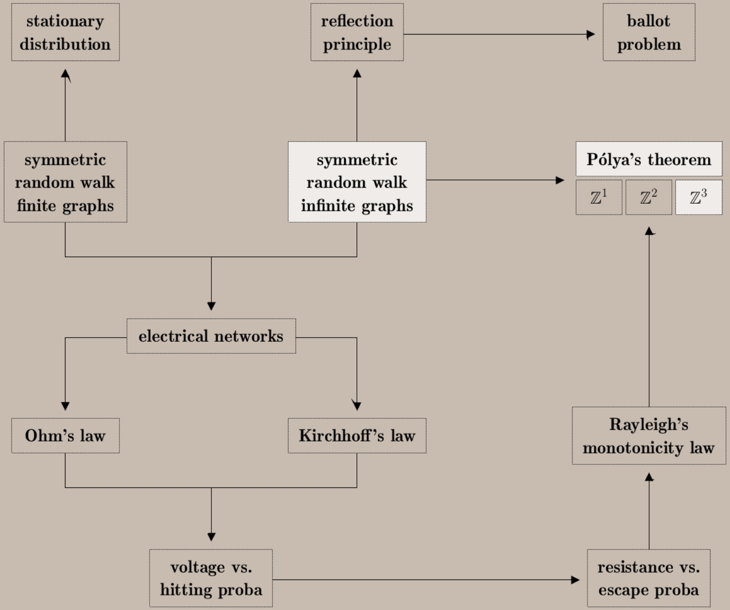


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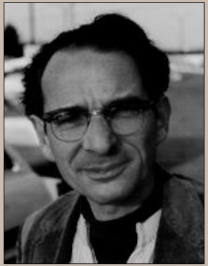
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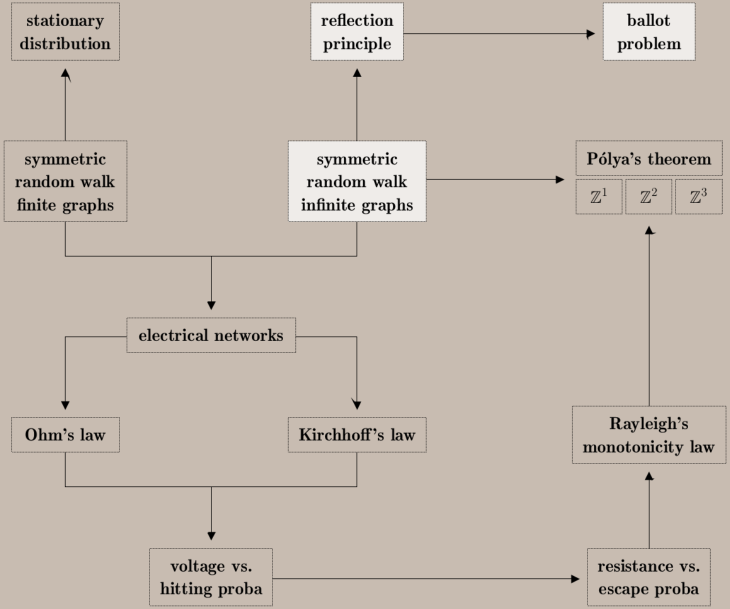

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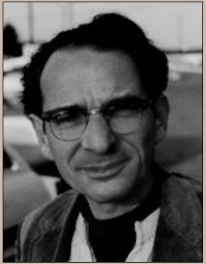

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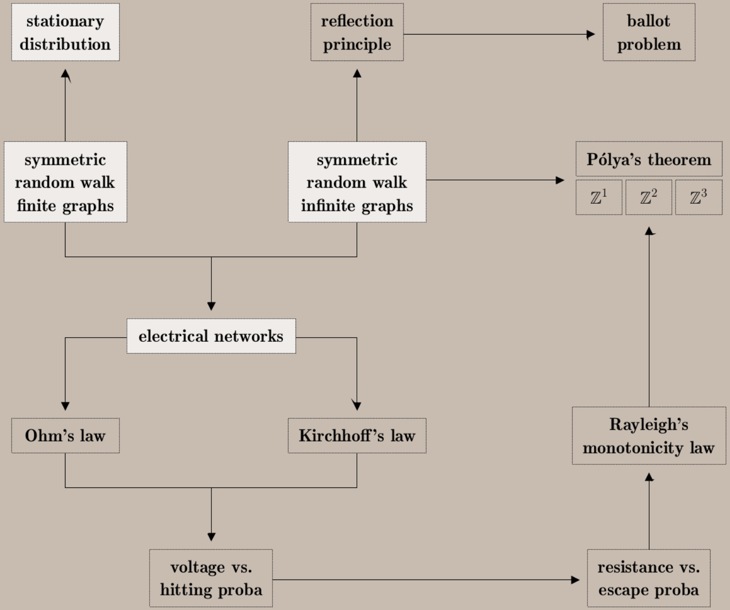

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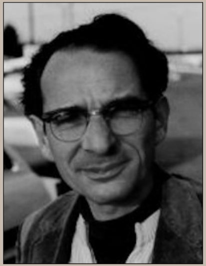

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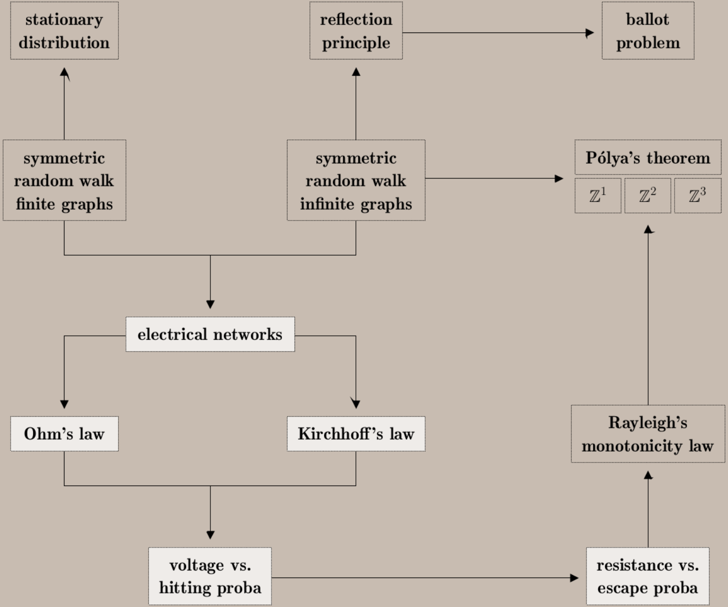
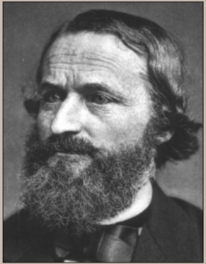
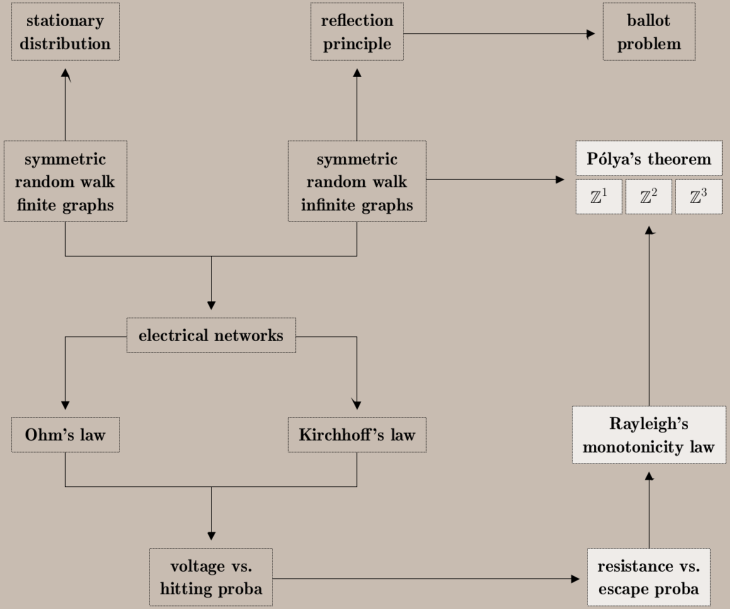



Georg Ohm


Gustav Kirchhoff



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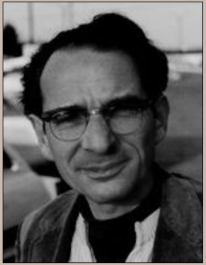

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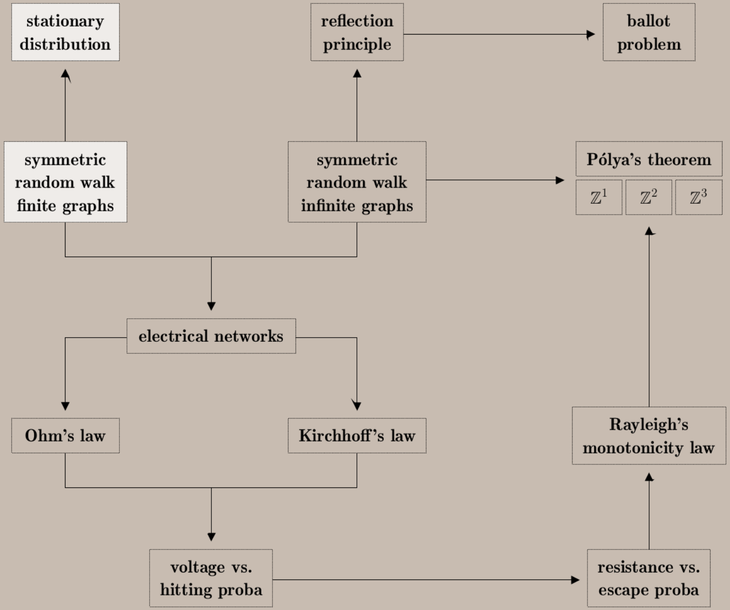

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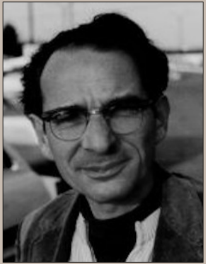
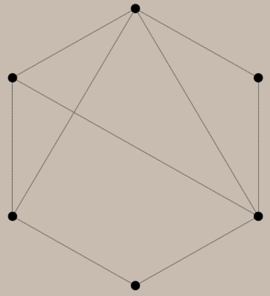
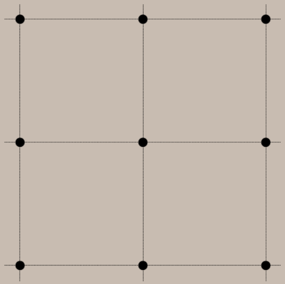


Frank Spitzer

Let $G = (V, E)$ = connected graph with bounded degree.

Symmetric random walk on G = process X_n that keeps track of the position of a particle moving on the vertex set and crossing each of the edges incident to its position with equal probability.





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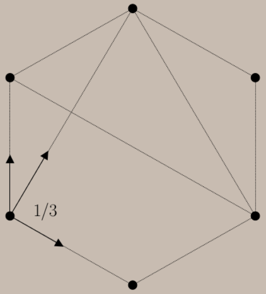

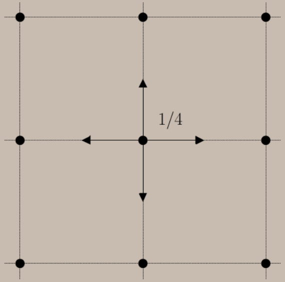

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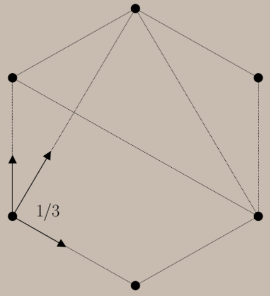

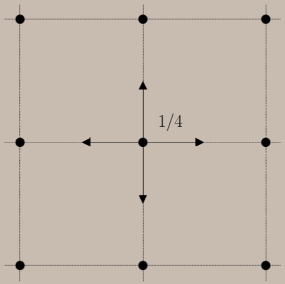

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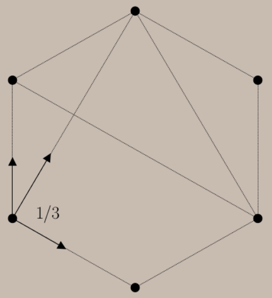

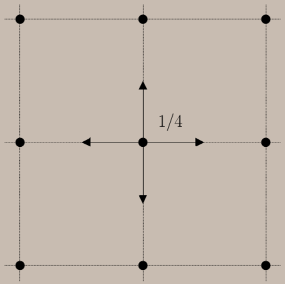

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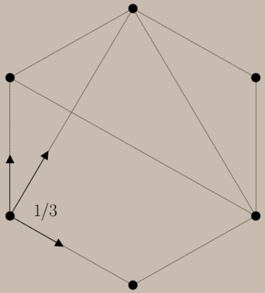

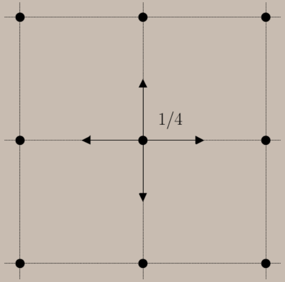

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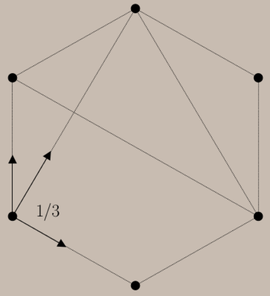
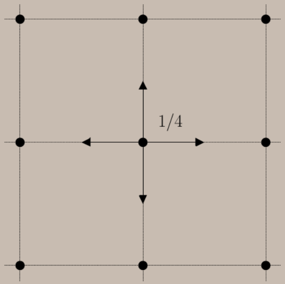
$\Longrightarrow$ process X_n irreducible and finite

$\Longrightarrow$ process X_n irreducible and positive recurrent





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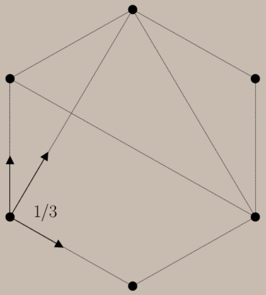
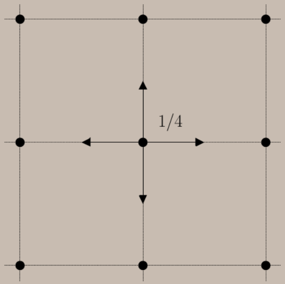
Assume G connected and finite

$\Longrightarrow$ process X_n irreducible and finite

$\Longrightarrow$ process X_n irreducible and positive recurrent

$\Longrightarrow$ unique stationary distribution π.





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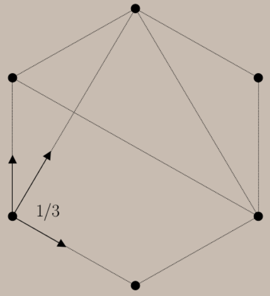
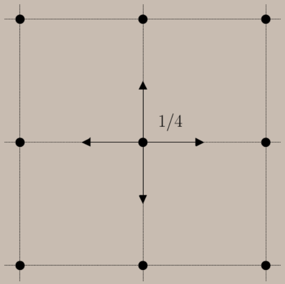
Solve $\pi P = \pi$ using the transition matrix?





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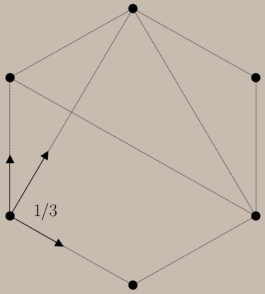
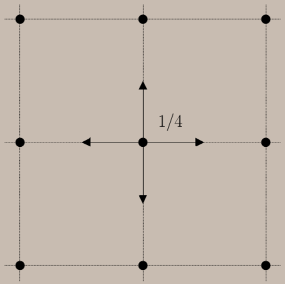
$$\deg(x)\, p(x, y) = \mathbf{1}\{(x, y) \in E\}$$





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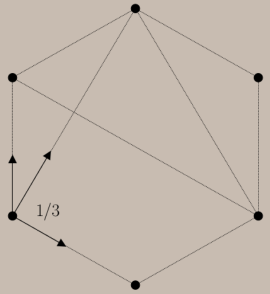

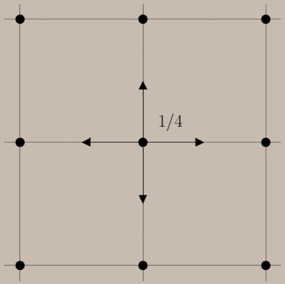

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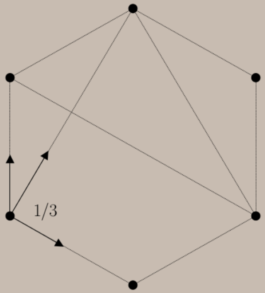
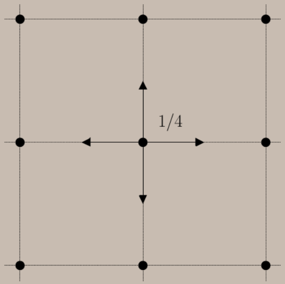
Reversibility $\Longrightarrow$ $\pi(x) = \deg(x) / \sum_{y \in V} \deg(y)$ for all $x \in V$.





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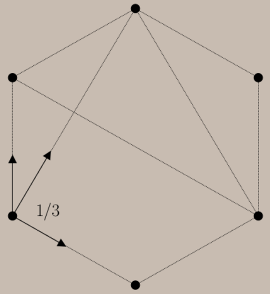
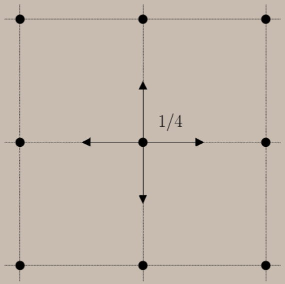
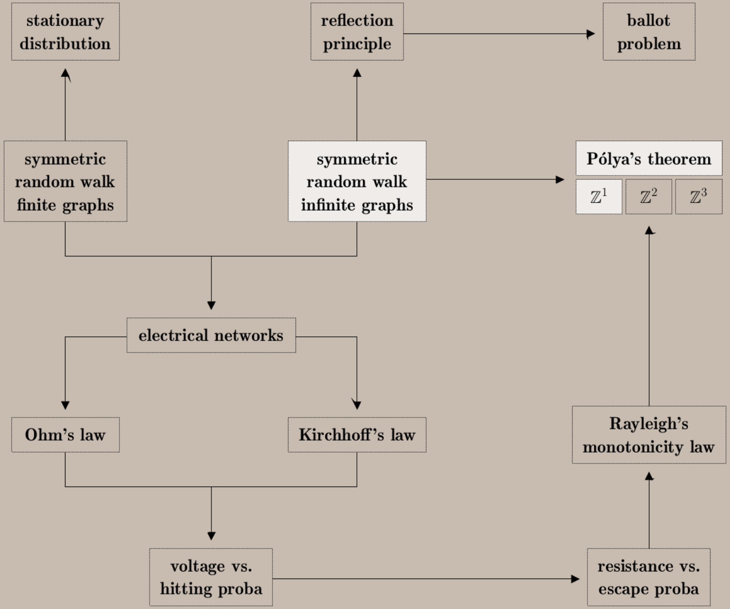
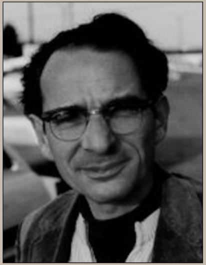
The fraction of time the symmetric random walk spends on each vertex is proportional to the degree of this vertex.



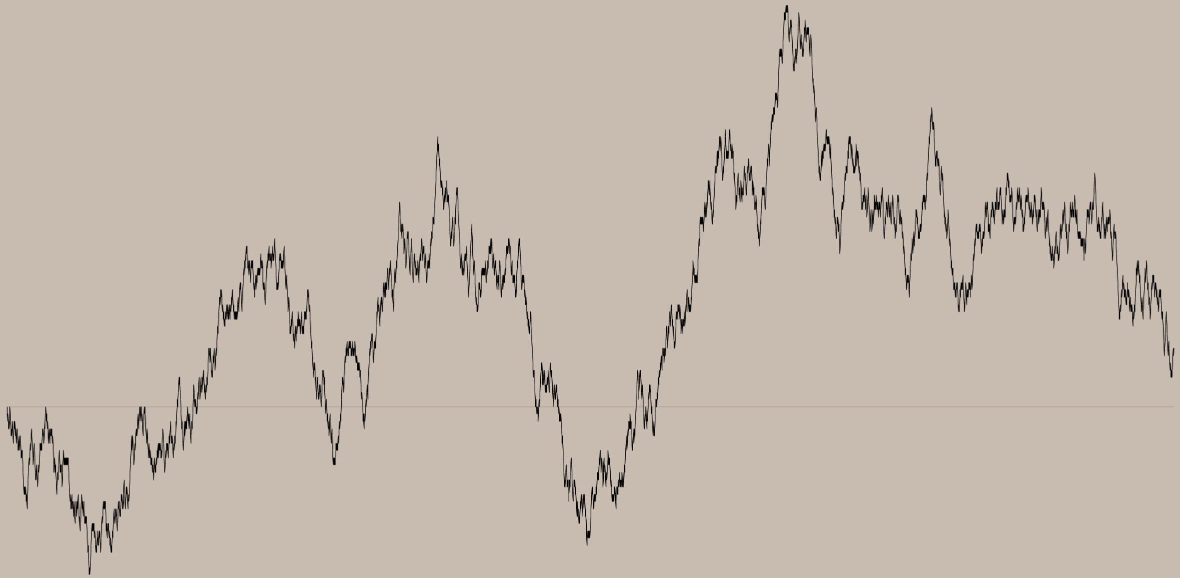


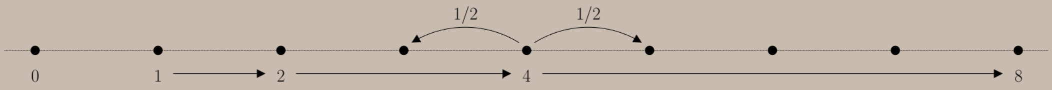


George Pólya

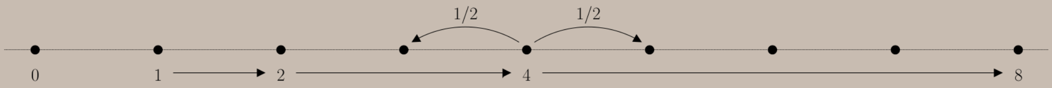


Frank Spitzer





Theorem. $\mathbb{Z}$ is recurrent/transient?



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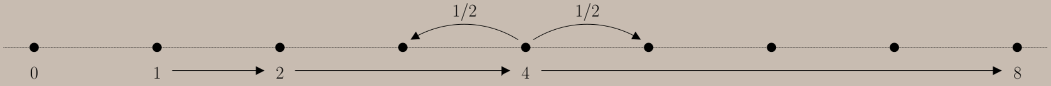
Lemma. State x is recurrent $\iff$ $\sum_n p_n(x,x) = \infty$.



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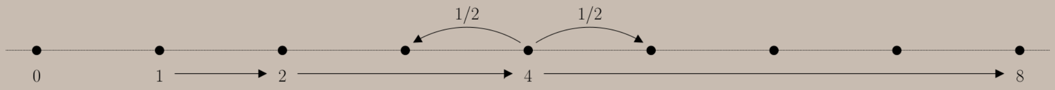
Proof 1. Period $= 2 \implies p_{2n}(0,0) = ?$



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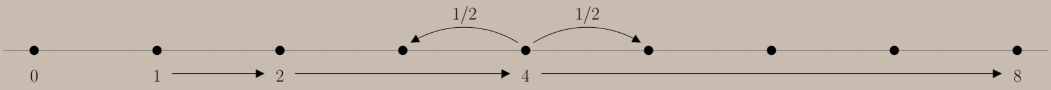

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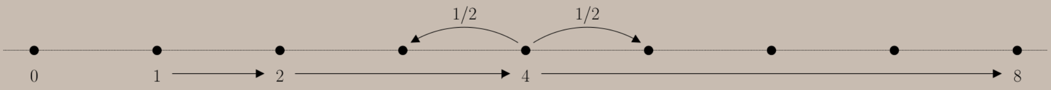

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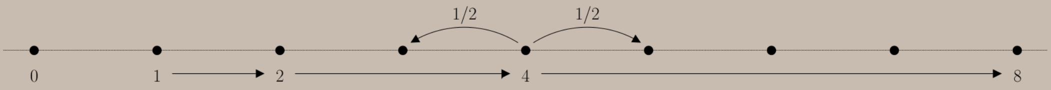
Stirling's formula: $n! \sim \sqrt{2\pi n}\,(n/e)^n \implies p_{2n}(0,0) \overset{\text{SF}}{\sim} \sqrt{1/\pi n}$.



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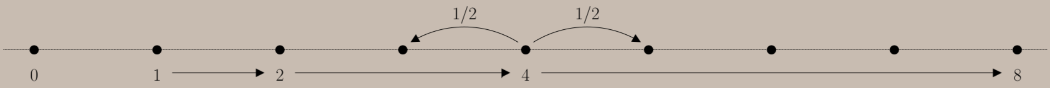

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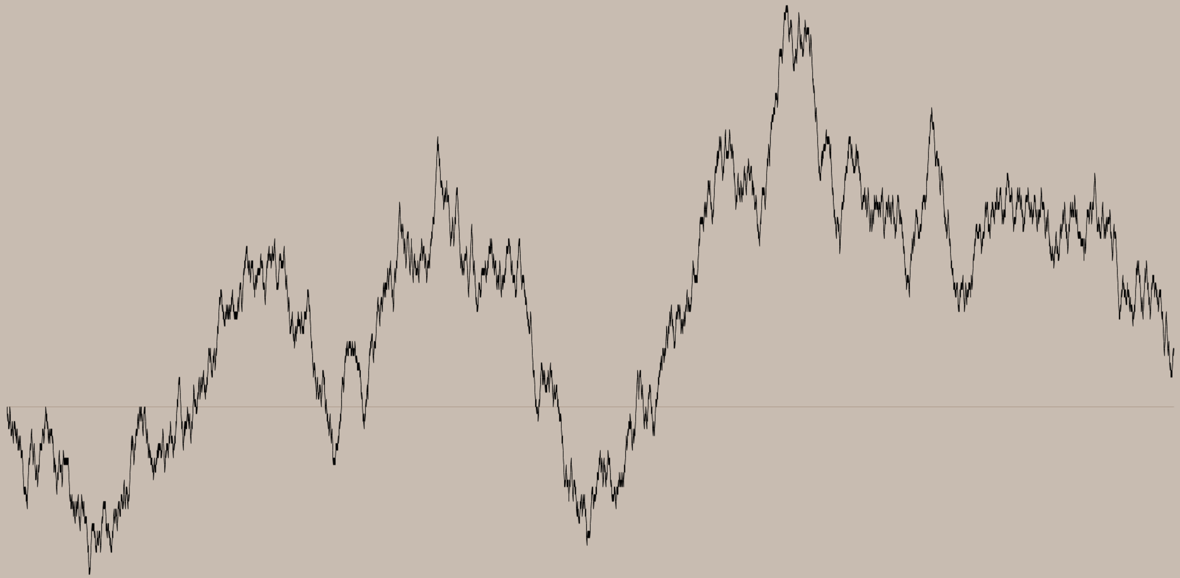

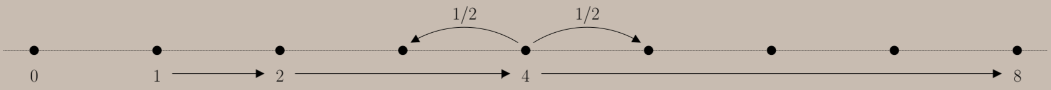

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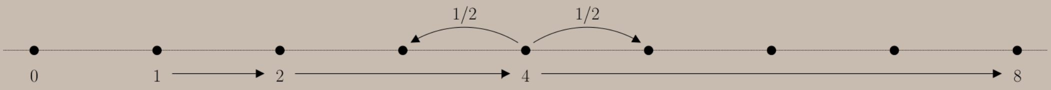

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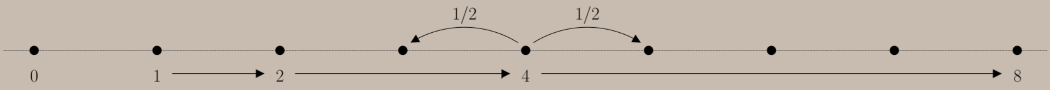

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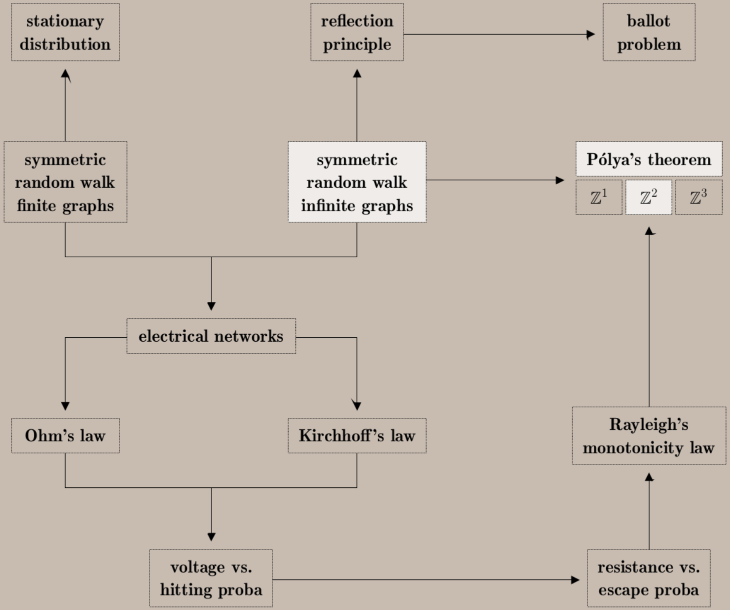
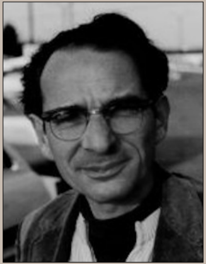



George Pólya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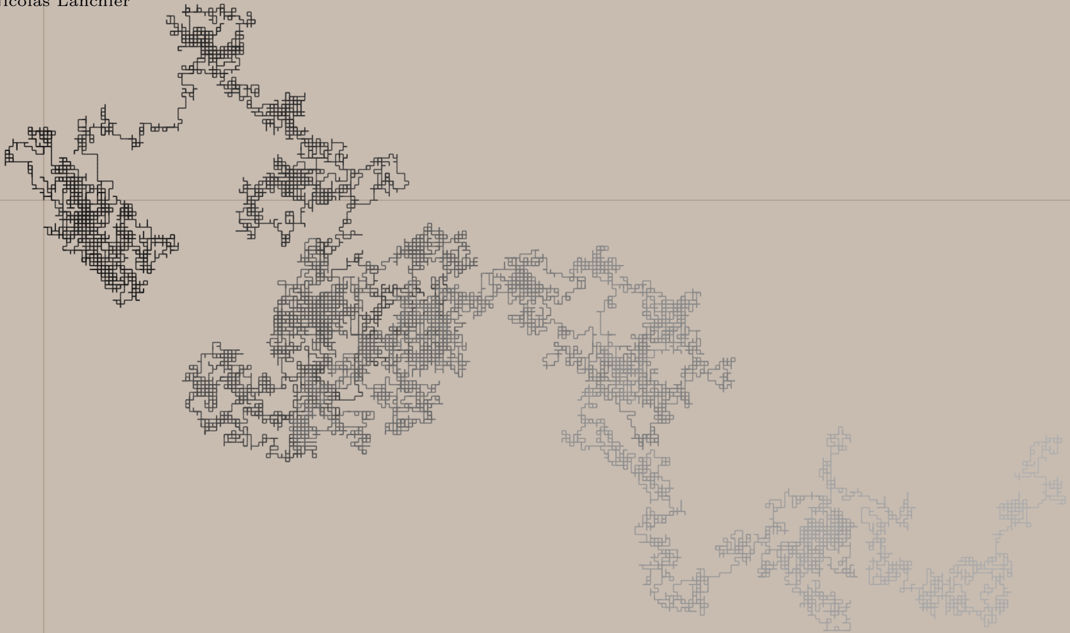

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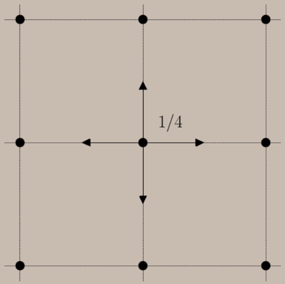

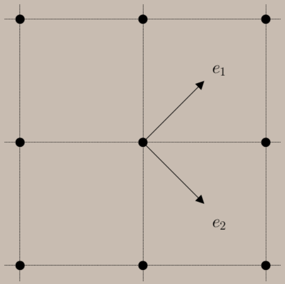

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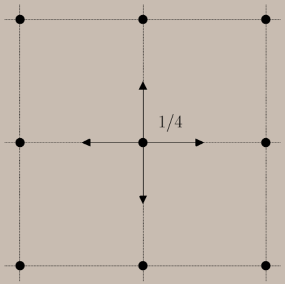

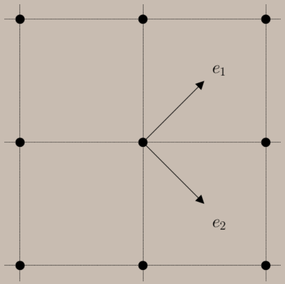

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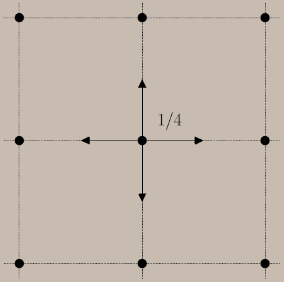

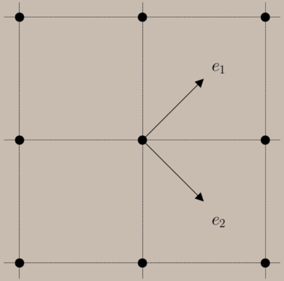

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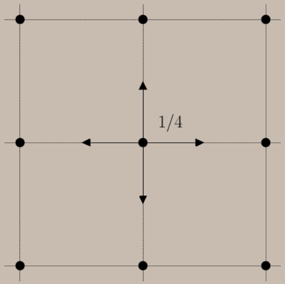

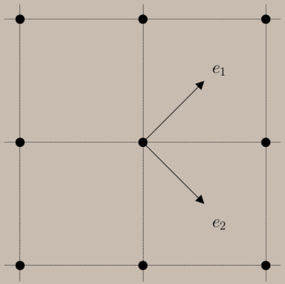

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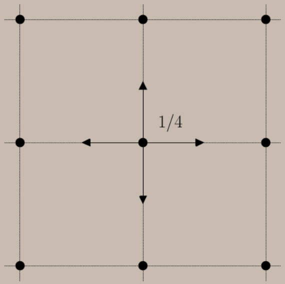

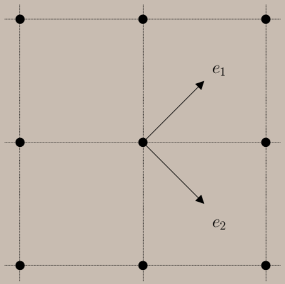

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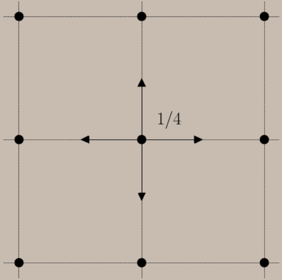

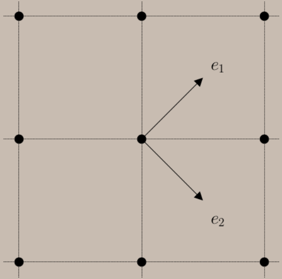

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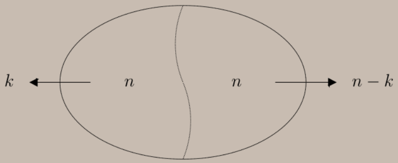

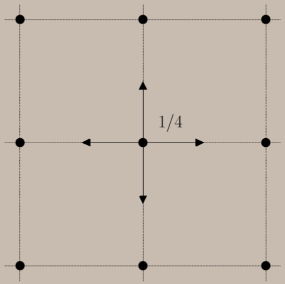

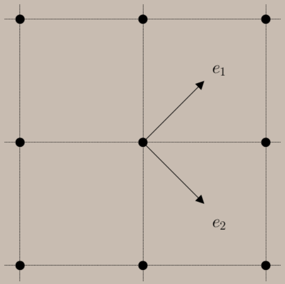

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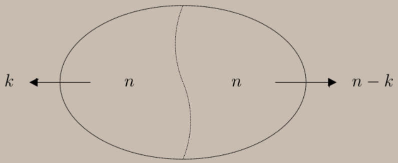

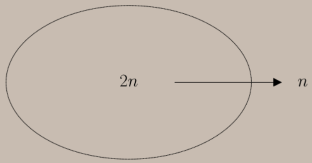

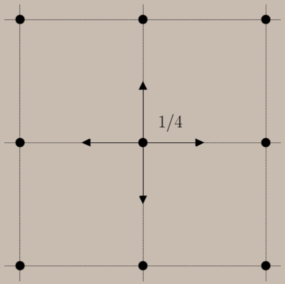

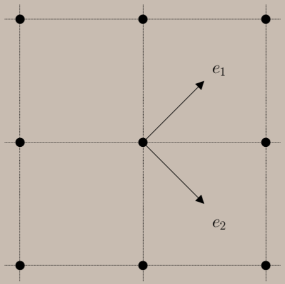

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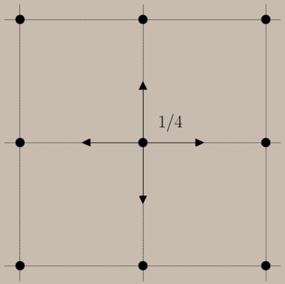

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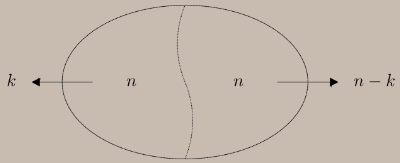

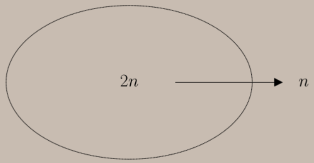

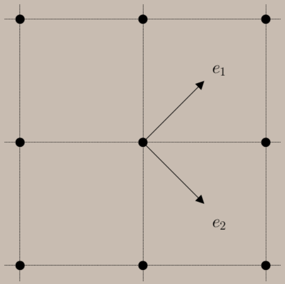

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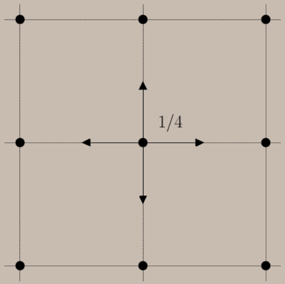

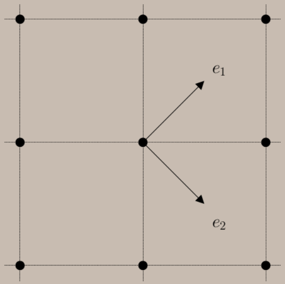

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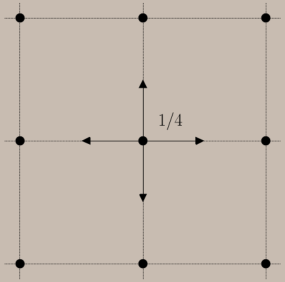
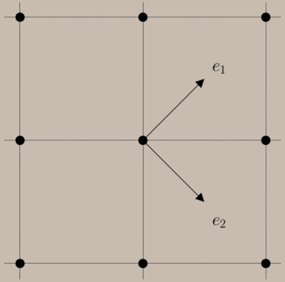
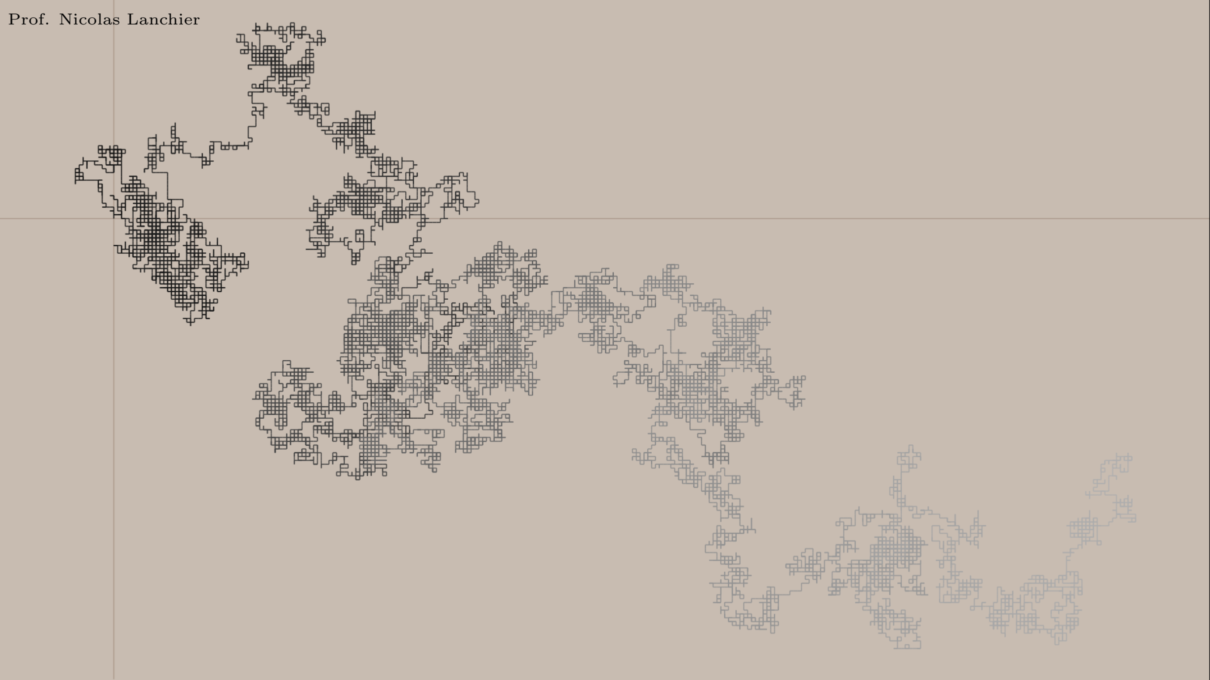




Prof. Nicolas Lanchier



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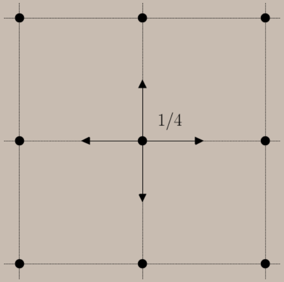

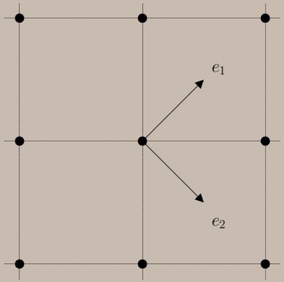

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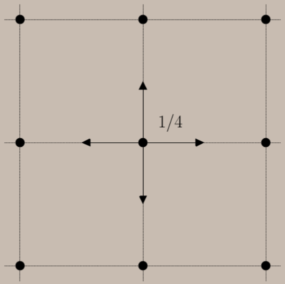

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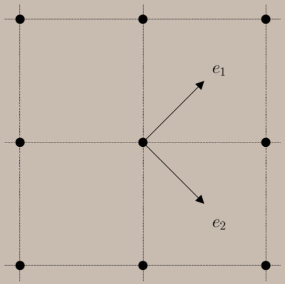
$$2n \;-\!\left[\begin{array}{lll} n & \rightarrow\uparrow & -\!\left[\begin{array}{cc} k & \rightarrow \\ n-k & \uparrow \end{array}\right. \\ & & \\ n & \leftarrow\downarrow & -\!\left[\begin{array}{cc} k & \leftarrow \\ n-k & \downarrow \end{array}\right. \end{array}\right.$$

$$\implies p_{2n}(0,0) = \textstyle\sum_{k=0}^n \binom{2n}{n}\binom{n}{k}^2 (1/4)^{2n}.$$

Counting subsets: $\sum_{k=0}^n \binom{n}{k}^2 = \sum_{k=0}^n \binom{n}{k}\binom{n}{n-k} = \binom{2n}{n}$

$$\implies p_{2n}(0,0) = \binom{2n}{n}^2 (1/4)^{2n} = (p_{2n}(0,0)\,|_{d=1})^2 \sim 1/\pi n$$

$\implies \mathbb{Z}^2$ is recurrent.



Proof 2. $Y_n, Z_n =$ independent symmetric random walks on $\mathbb{Z}$

$\implies X_n = Y_n\, e_1 + Z_n\, e_2 =$ symmetric random walk on $\mathbb{Z}^2$.

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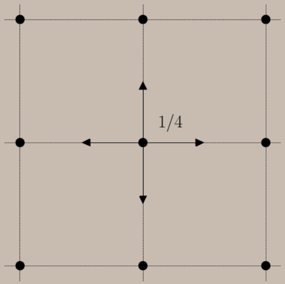

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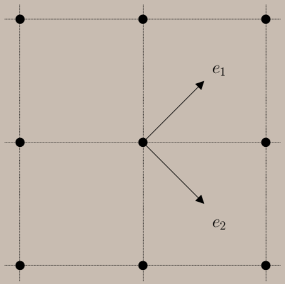

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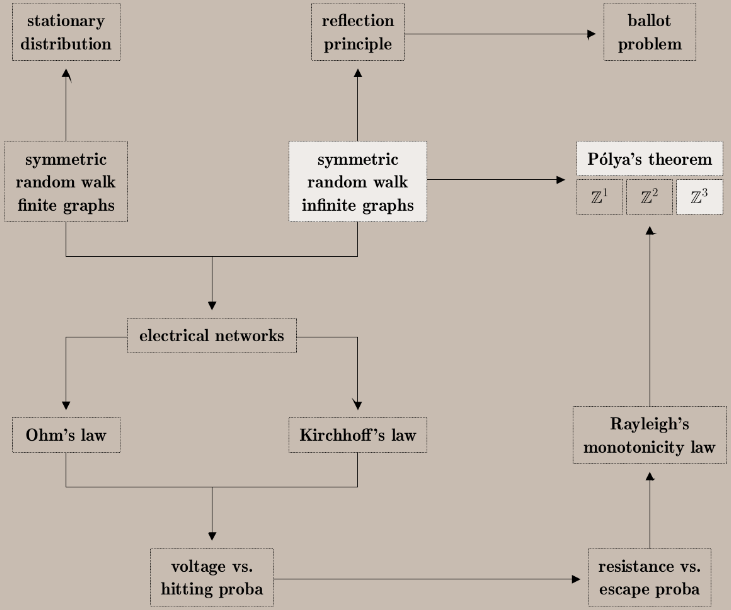

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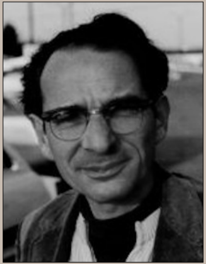
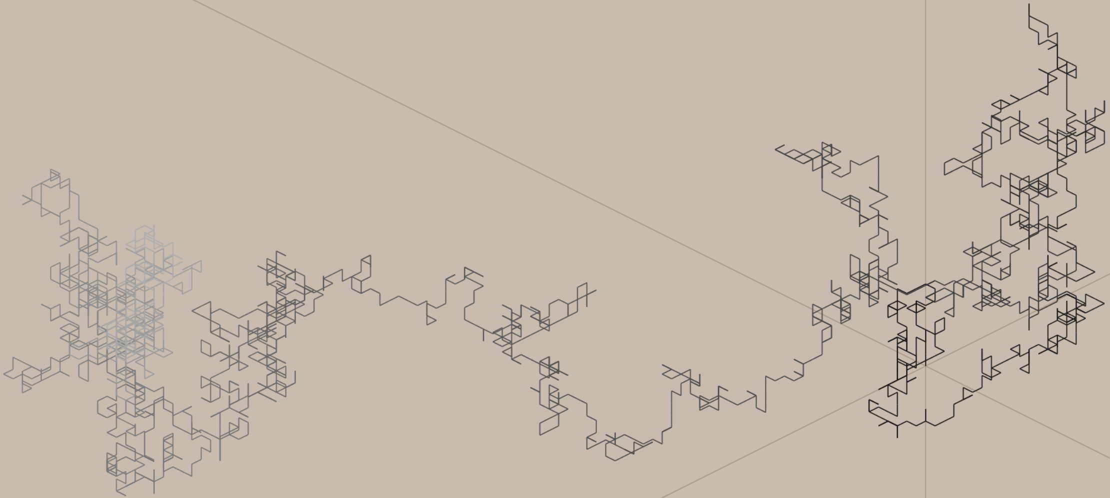
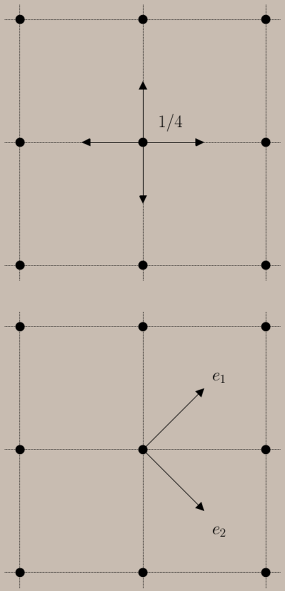


Frank Spitzer

Prof. Nicolas Lanchier



Theorem. $\mathbb{Z}^3$ is recurrent/transient?



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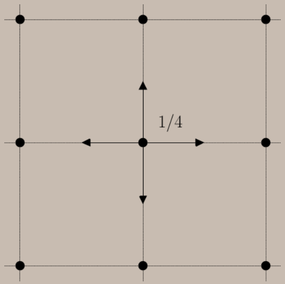

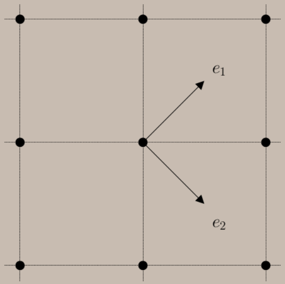

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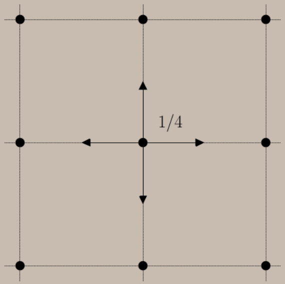

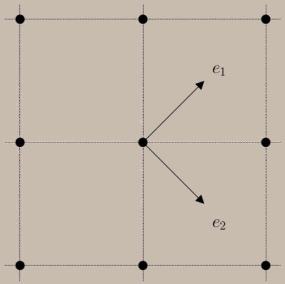

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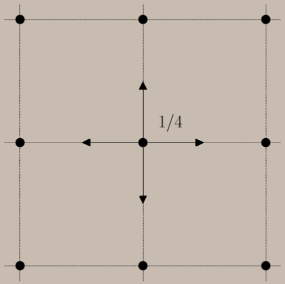

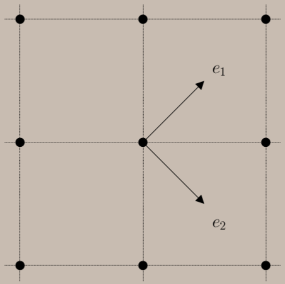

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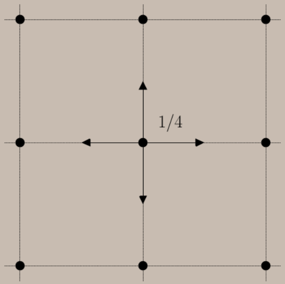

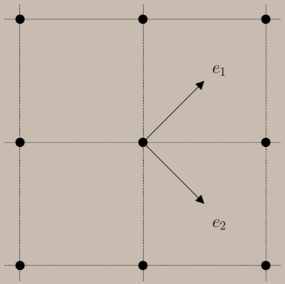

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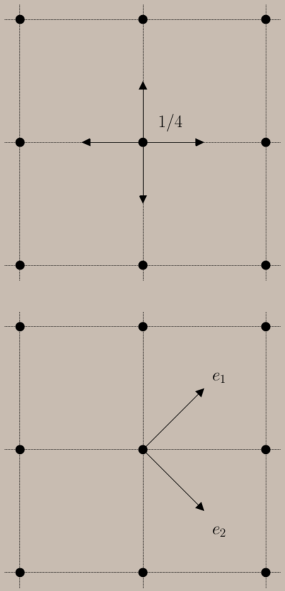

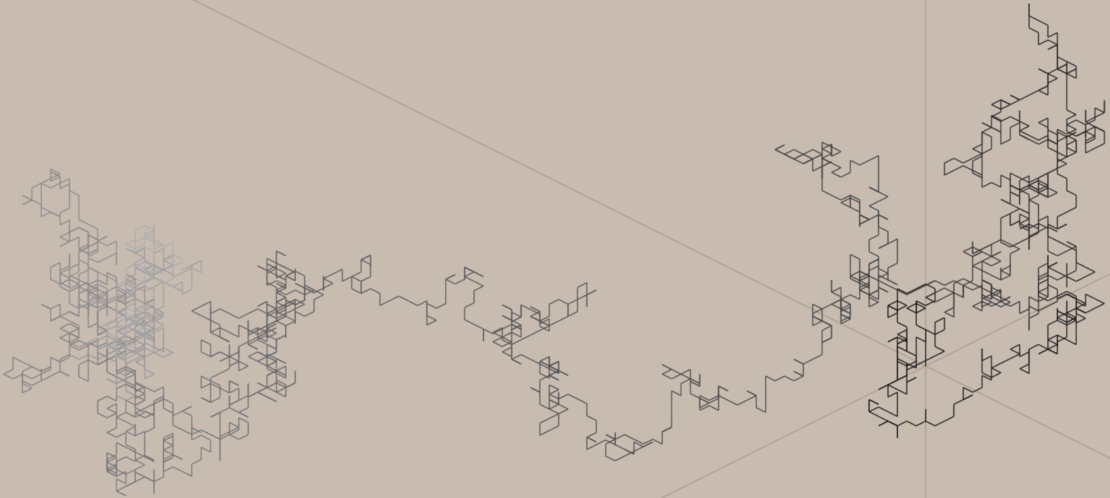

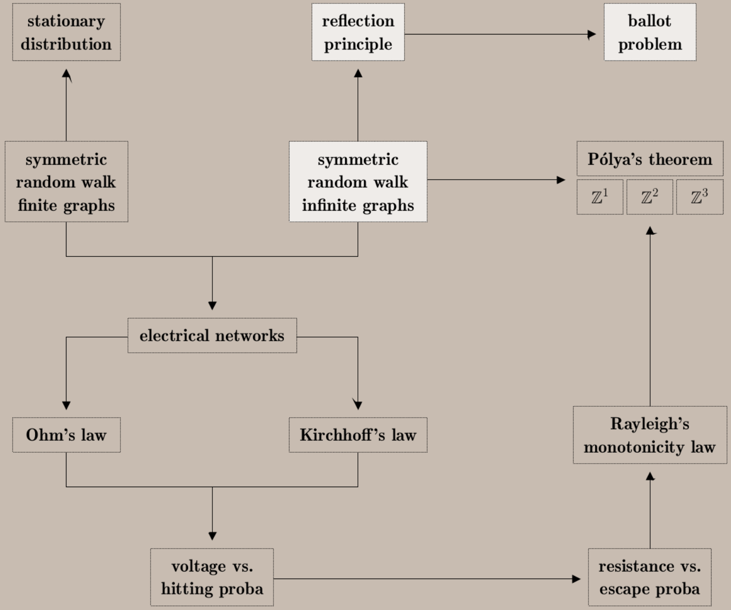


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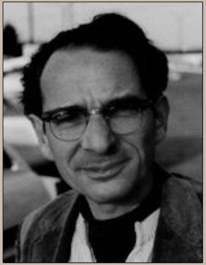
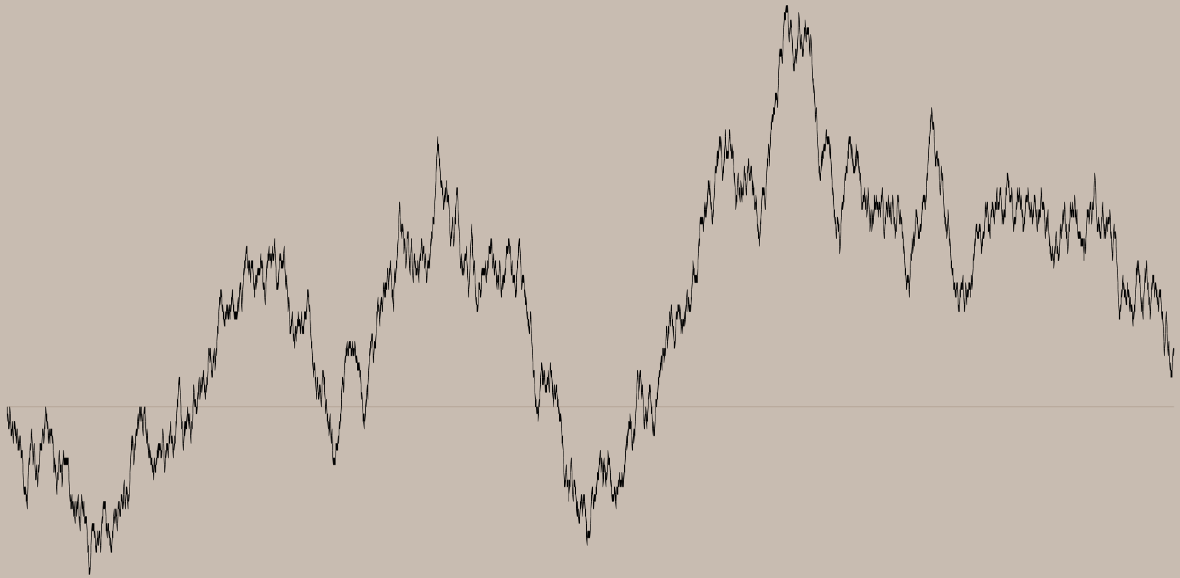

Frank Spitzer



Let X_n = symmetric random walk on $\mathbb{Z}$.

$$N_n(x,y) \;=\; \#\text{ paths } (0,x) \to (n,y).$$

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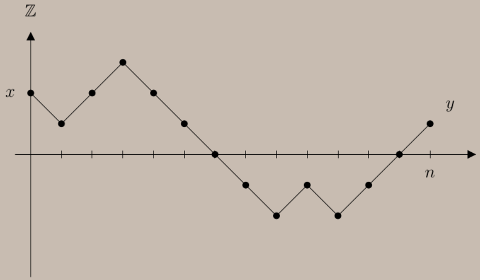

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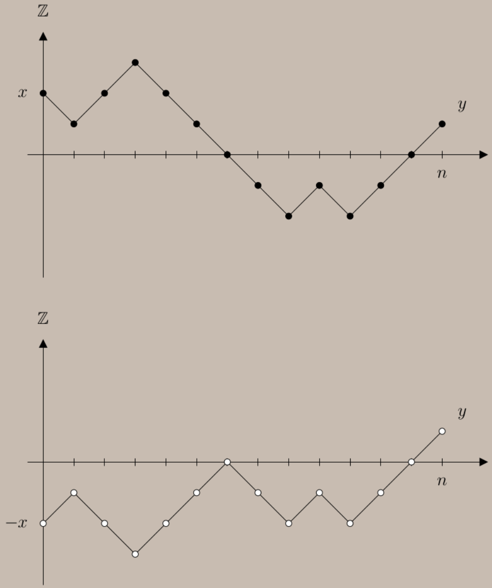
Reflection principle:

$$N_n^0(x,y) = N_n(-x,y) \text{ for all } x,y > 0.$$



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Proof. Reflecting the part of the path before the first visit to 0 across the x-axis defines a bijection between the two sets

$$A \ = \ \text{set of paths } (0,x) \to (n,y) \text{ that visit } 0$$

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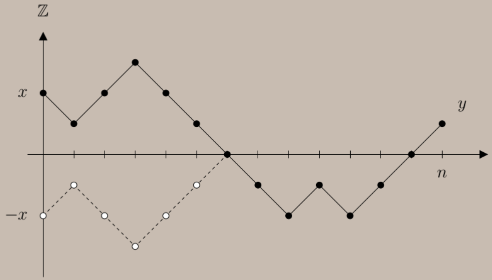

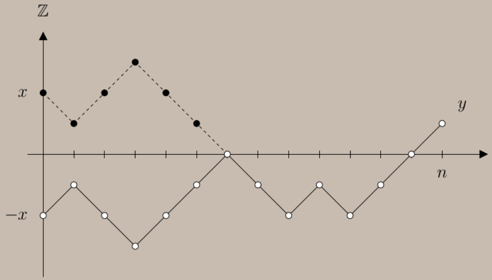

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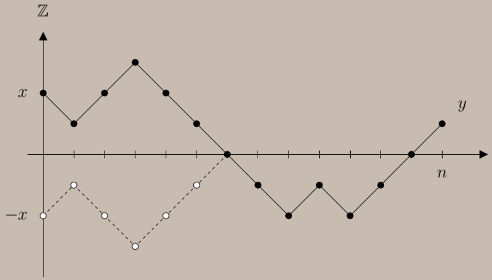

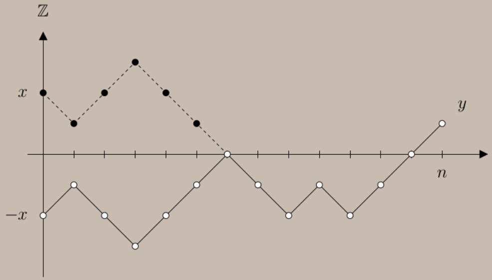

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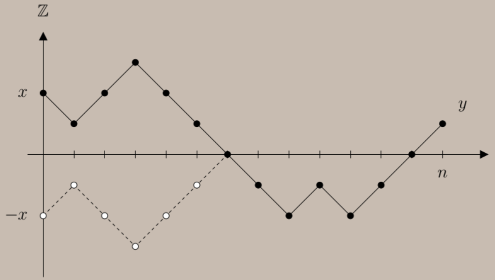

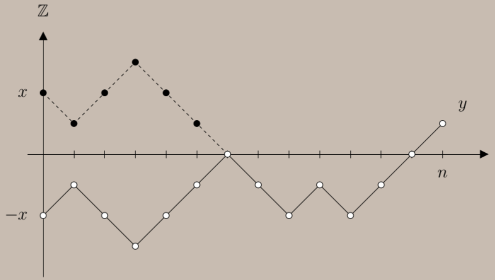

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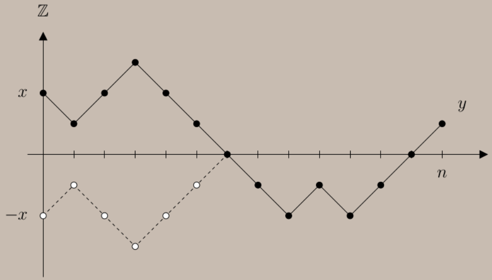

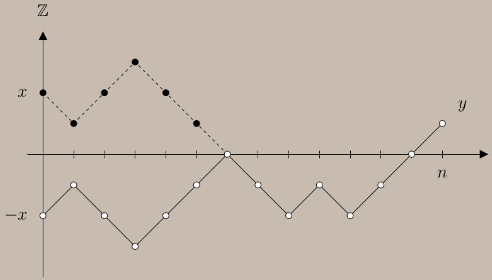

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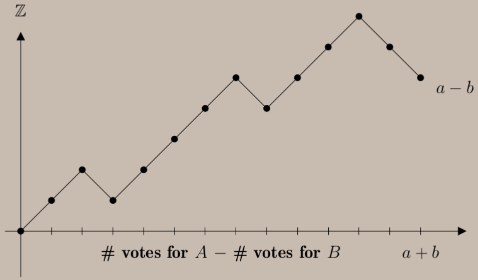

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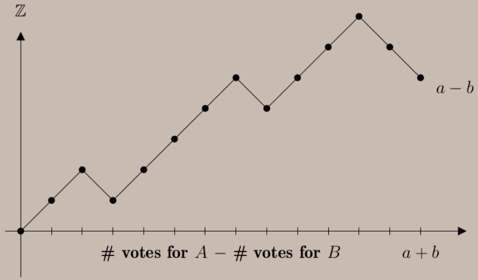

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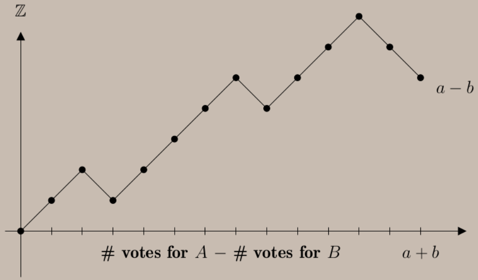

First move is $\uparrow \implies$ # paths with no return to 0 is

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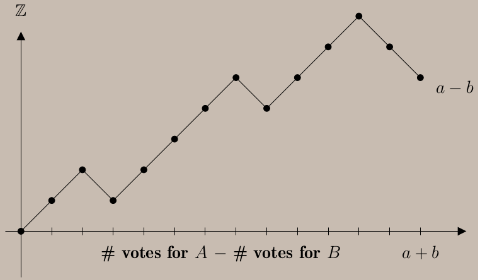

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Counting paths (permutation with repetition):

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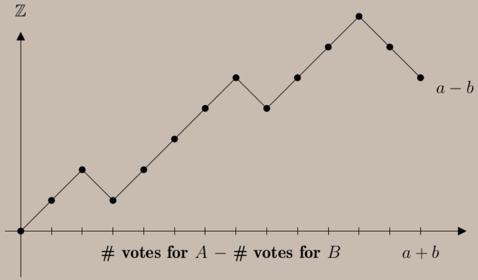


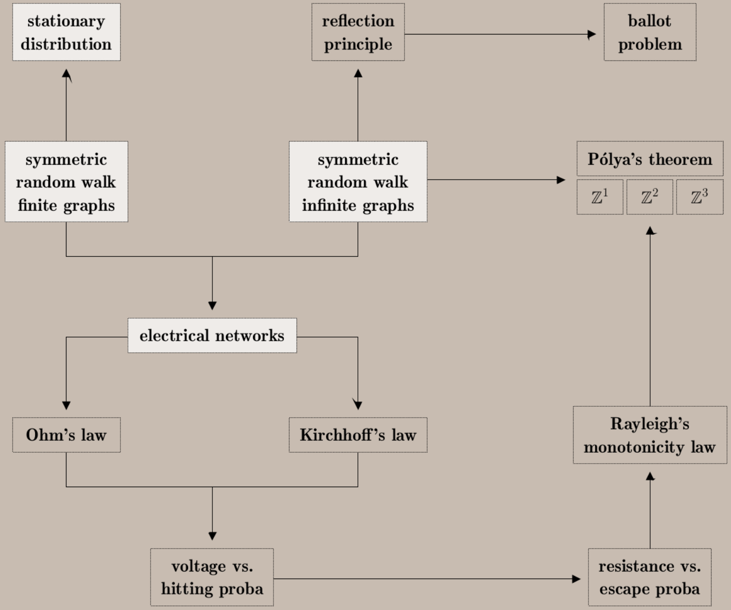
First move is $\uparrow \implies$ # paths with no return to 0 is

$$N_{a+b-1}(1,a-b) - N_{a+b-1}^0(1,a-b) \overset{\text{RP}}{=} N_{a+b-1}(1,a-b) - N_{a+b-1}(-1,a-b).$$

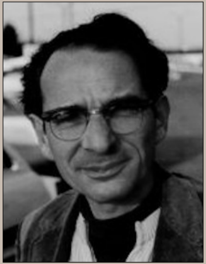
Counting paths (permutation with repetition):

$$N_{a+b-1}(1,a-b) = (a-1\uparrow, b\downarrow) = \binom{a+b-1}{b} \quad \text{and} \quad N_{a+b-1}(-1,a-b) = (a\uparrow, b-1\downarrow) = \binom{a+b-1}{a}.$$

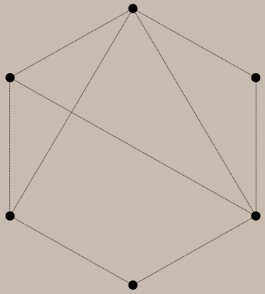
$$\implies p_A = (N_{a+b-1}(1,a-b) - N_{a+b-1}(-1,a-b))/N_{a+b}(0,a-b)$$
$$= \left[\binom{a+b-1}{b} - \binom{a+b-1}{a}\right] / \binom{a+b}{a} = (a-b)/(a+b).$$

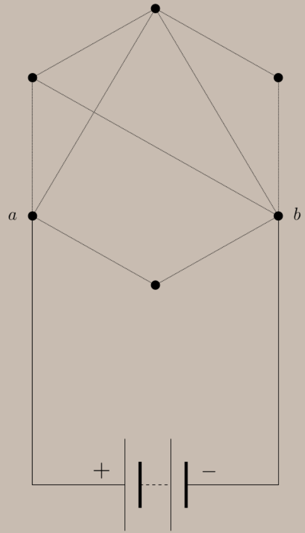


George Pólya

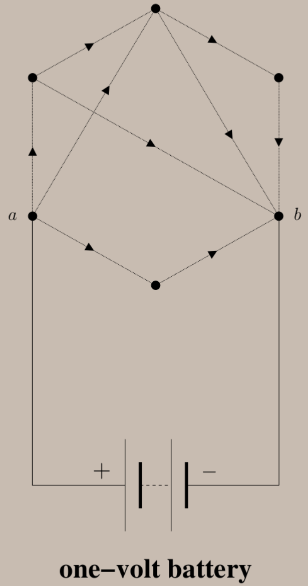


Frank Spitzer



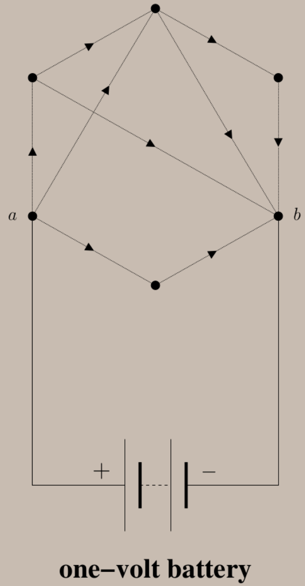


one–volt battery


a
b
+
−
one–volt battery

Electrical network. Equip $G = (V, E)$ with a function

$$c : E \to [0, \infty] = \text{conductance along the edges.}$$



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Random walk on (G, c) = Markov chain X_n with transitions

$$p(x, y) = c(x, y)\, \mathbf{1}\{(x, y) \in E\} / c(x)$$

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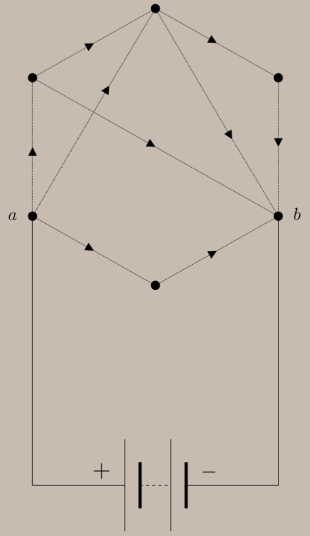

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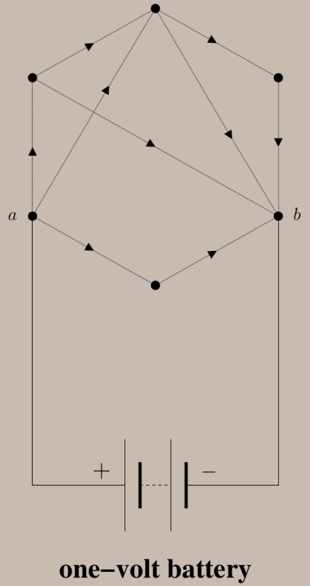

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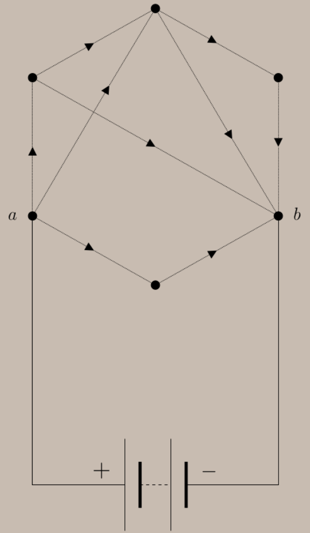
Assume G connected and finite, and $c \in (0, \infty)$ on E



one–volt battery

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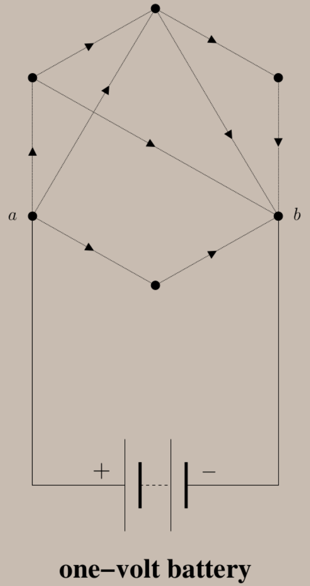
$\Longrightarrow$ process X_n irreducible and finite



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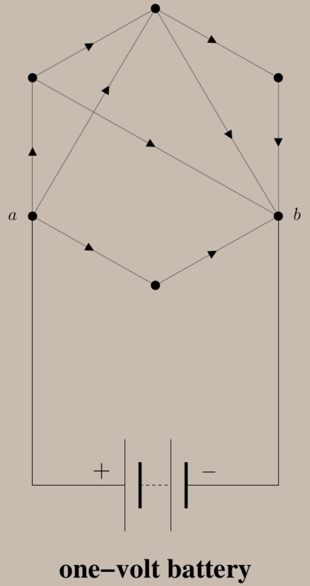
$\implies$ process X_n irreducible and finite

$\implies$ process X_n irreducible and positive recurrent



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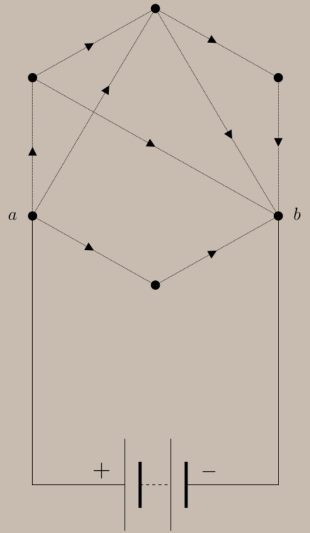
Assume G connected and finite, and $c \in (0, \infty)$ on E

$\Longrightarrow$ process X_n irreducible and finite

$\Longrightarrow$ process X_n irreducible and positive recurrent

$\Longrightarrow$ unique stationary distribution π.



one–volt battery

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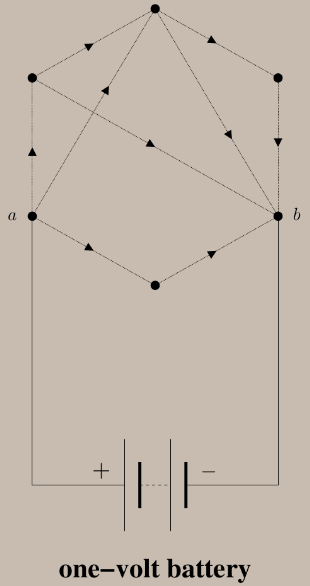
Solve $\pi P = \pi$ using the transition matrix?



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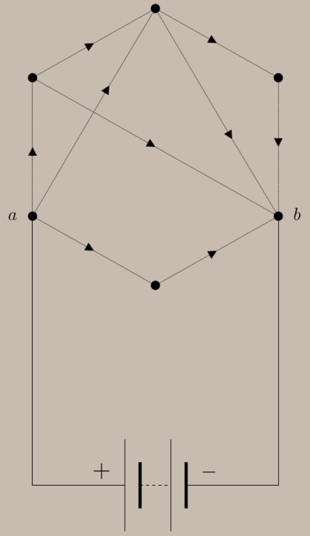

one–volt battery

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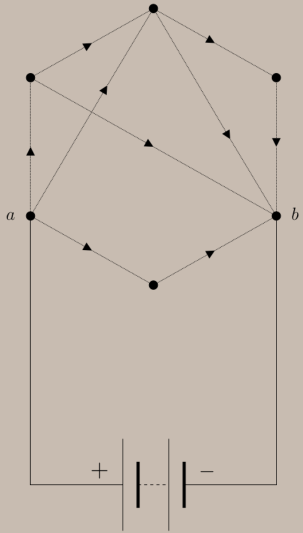
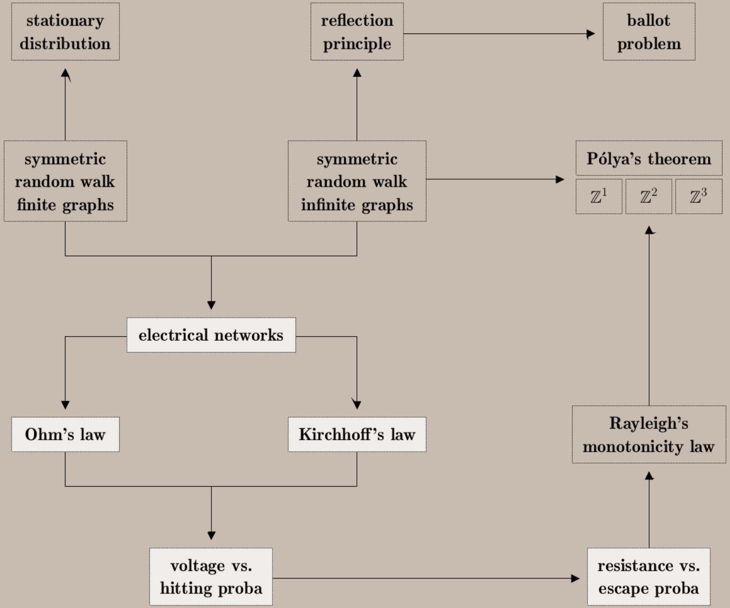
$$c(x)\, p(x, y) = c(x, y)\, \mathbf{1}\{(x, y) \in E\} = c(y)\, p(y, x).$$

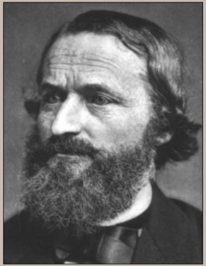
Reversibility $\Longrightarrow$ $\pi(x) = c(x) / \sum_{y \in V} c(y)$ for all $x \in V$.



one−volt battery



Georg Ohm



Gustav Kirchhoff

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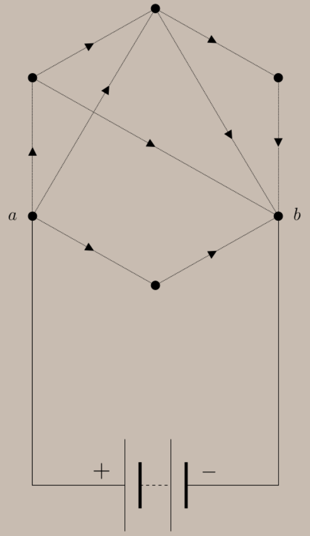
Plug a one-volt battery from vertex a to vertex b.



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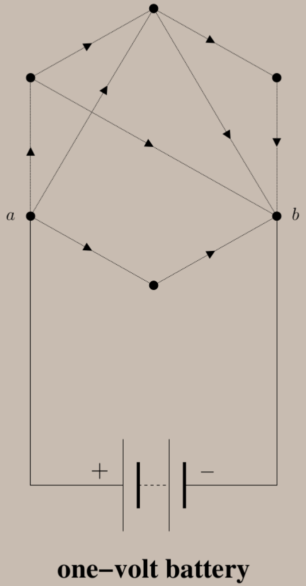

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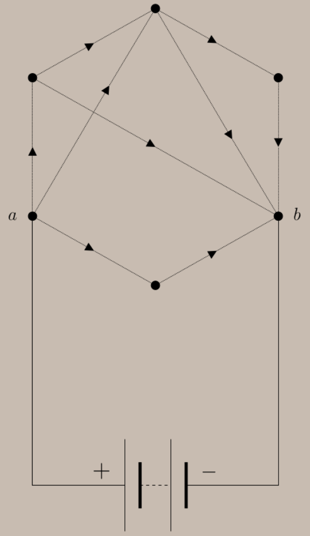
Kirchhoff's law: The total current flowing through any vertex other than a and b is equal to zero:

$$i(x) = \sum_{z \sim x} i(x, z) = 0 \quad \text{for all} \quad x \neq a, b.$$



one–volt battery

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Theorem. Let $\tau_y = \inf\{n : X_n = y\}$. Then,

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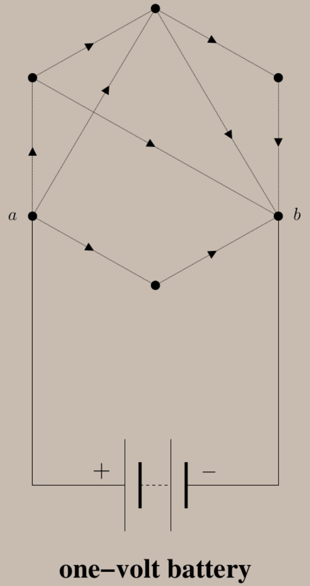

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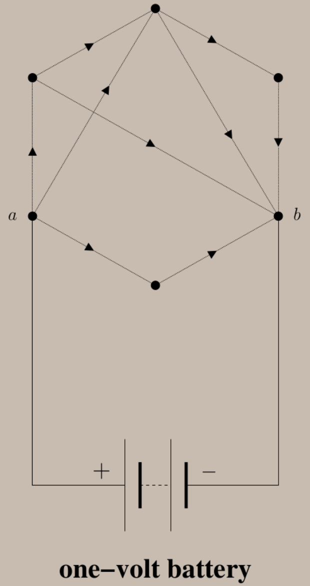
Proof. Prove that v and h are both harmonic.



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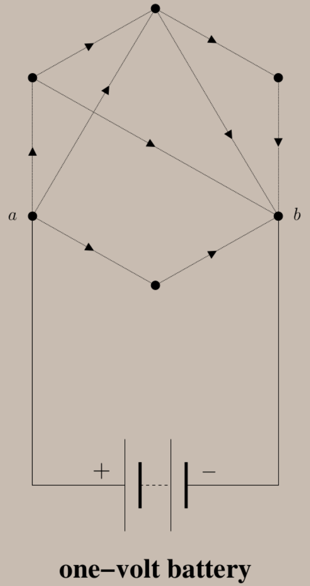

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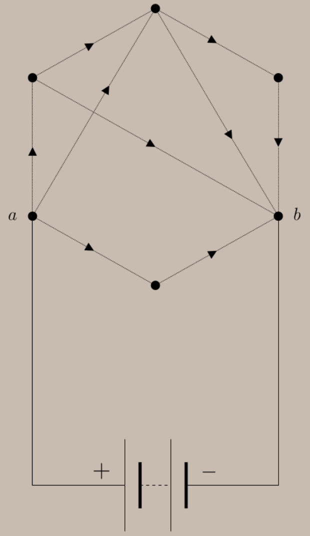

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Hitting probability: For all $x \neq a, b$,

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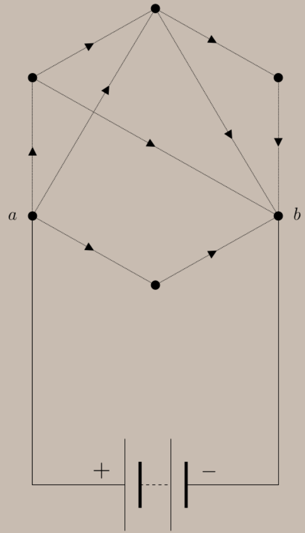

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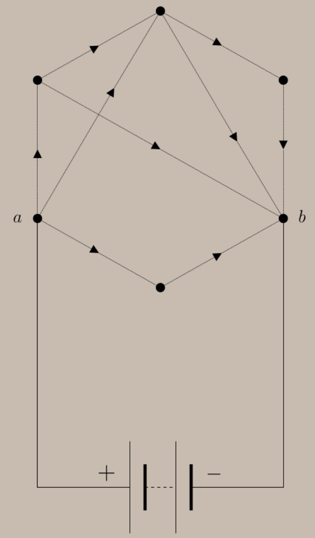
$\Longrightarrow$ v and h **harmonic** with $v(a) = h(a) = 1$ and $v(b) = h(b) = 0$.



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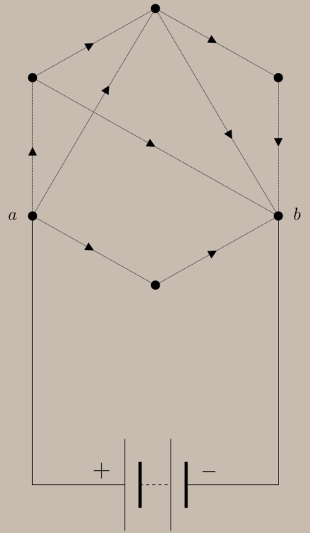
$\Longrightarrow$ v and h harmonic with $v(a) = h(a) = 1$ and $v(b) = h(b) = 0$.

Maximum principle $\Longrightarrow$ $v \equiv h$.



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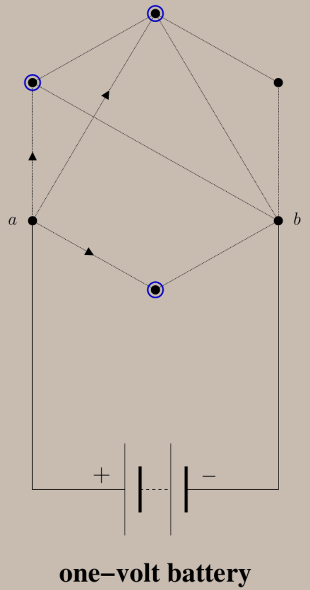
Definition. Effective resistance between a and b

$$R_{\mathit{eff}}(a,b) = 1 / \sum_{z \sim a} i(a,z).$$



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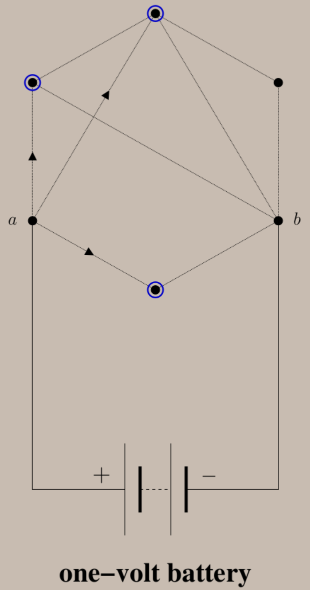

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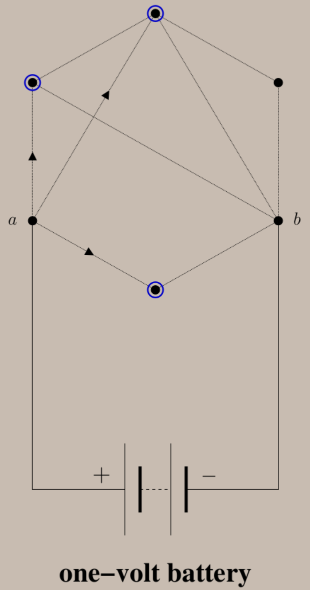

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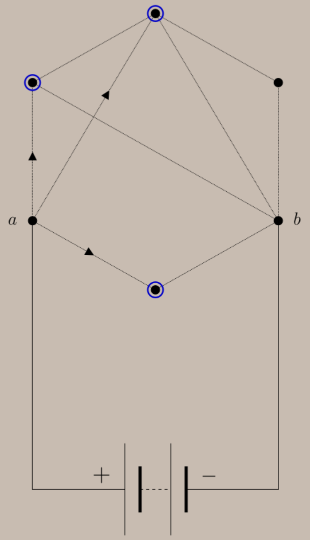

one–volt battery

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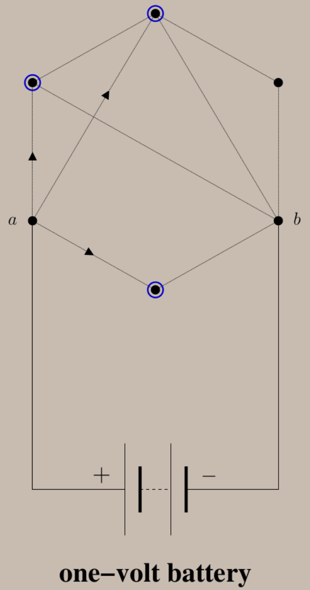

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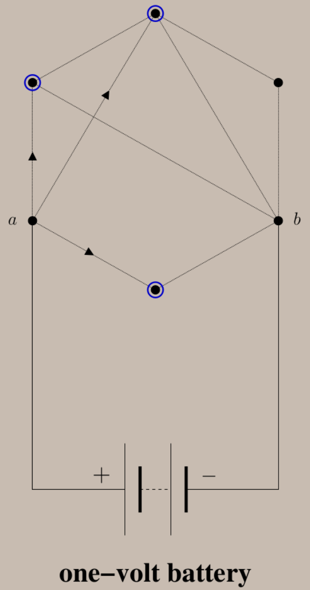
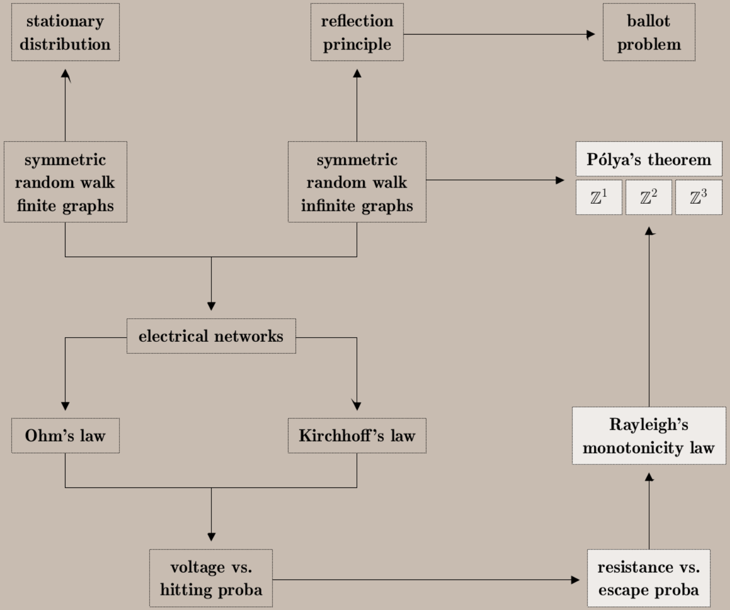
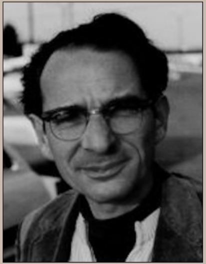




George Pólya



Frank Spitzer

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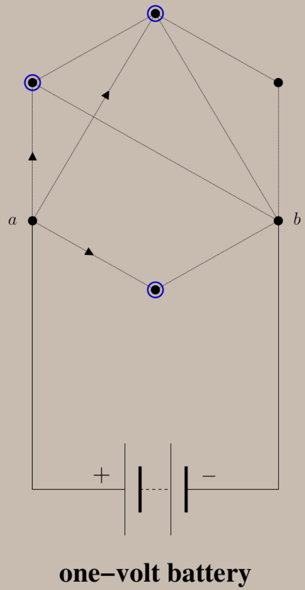

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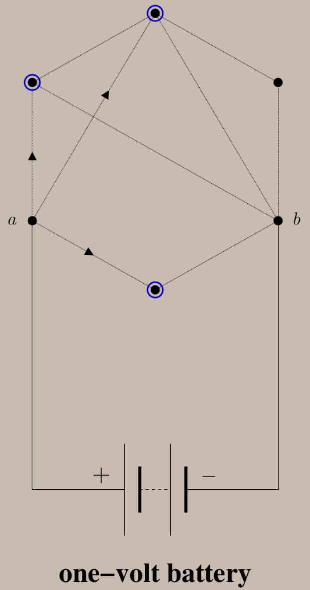

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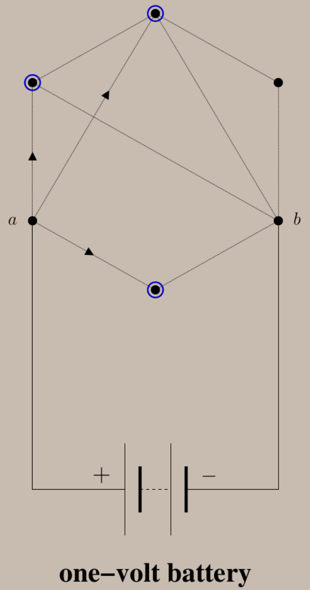

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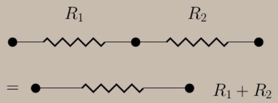

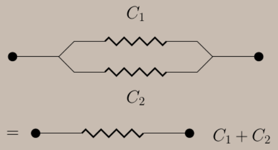

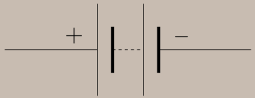

one–volt battery

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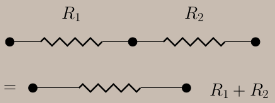
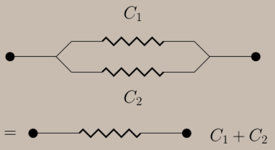
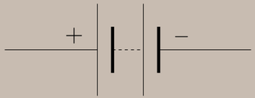
Rayleigh's monotonicity law: Increasing/decreasing the resistance of an edge of the network can only increase/decrease the effective resistance between any two vertices.







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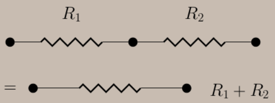

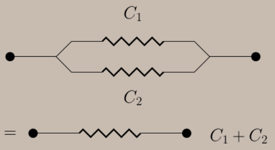

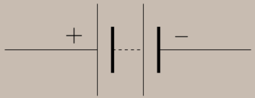

one–volt battery

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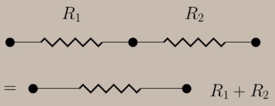

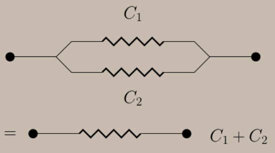

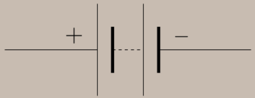

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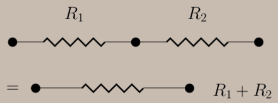

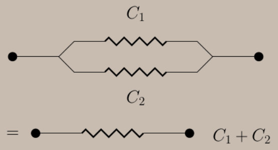

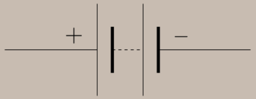

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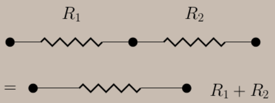
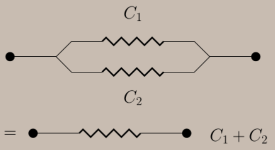
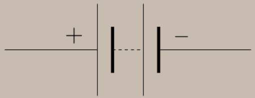
$\Longrightarrow$ Any subgraph of a recurrent graph is recurrent.

Any supergraph of a transient graph is transient.

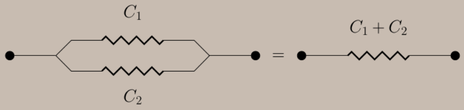






one–volt battery


R_1
R_2
$=$
$R_1 + R_2$


C_1
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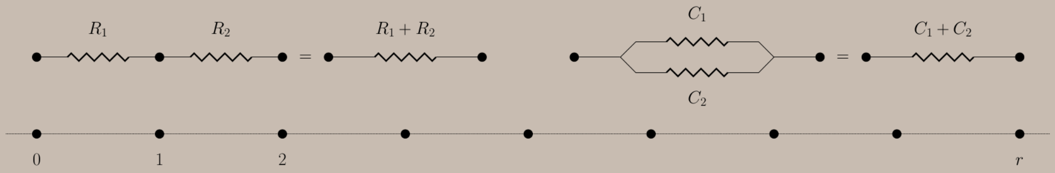
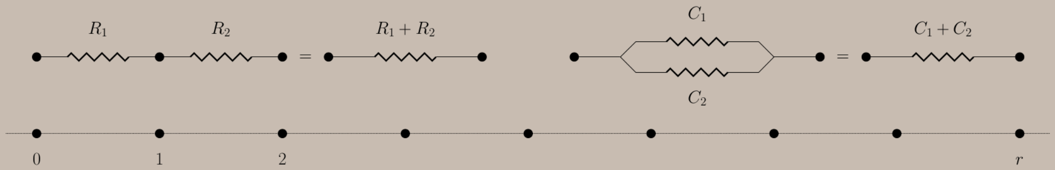


Theorem. (G, c) is recurrent $\iff R_{eff}(a, \infty) = \infty$.



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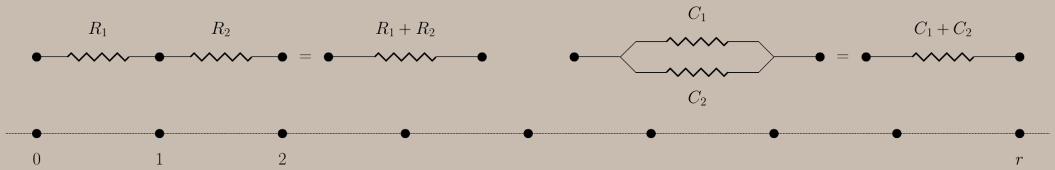
Theorem. $\mathbb{Z}$ is recurrent.



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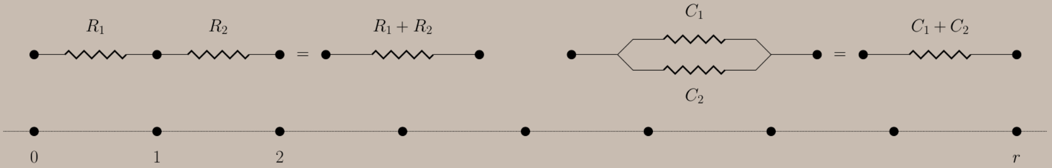

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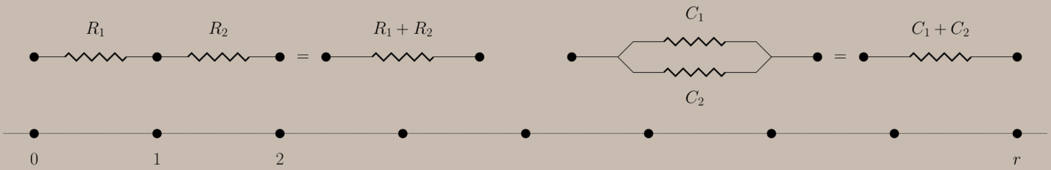

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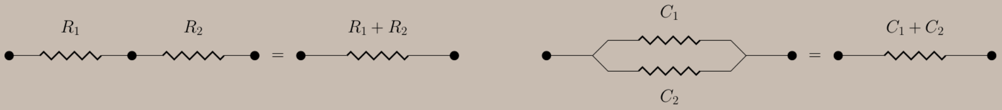
Proof. Shorting = identifying all the vertices on the same square centered at the origin



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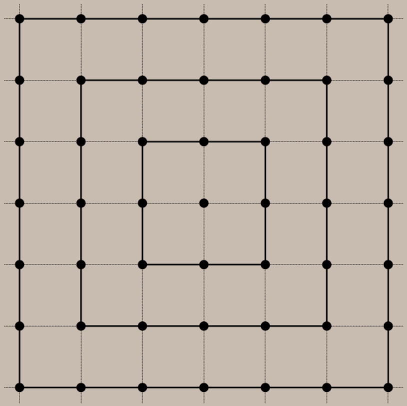

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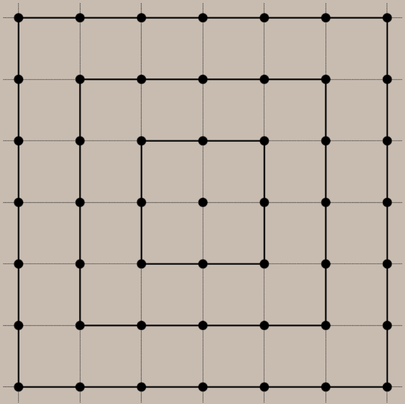
Theorem. $\mathbb{Z}^2$ is recurrent.

Proof. Shorting = identifying all the vertices on the same square centered at the origin $\Longrightarrow$

$$R_{\mathit{eff}}(0, r) \geq \frac{1}{4} + \frac{1}{12} + \frac{1}{20} + \cdots + \frac{1}{8r - 4} \to \infty$$

as the radius $r \to \infty$.



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Proof. Cutting = find an infinite subtree embedded in $\mathbb{Z}^3$ with a finite effective resistance.

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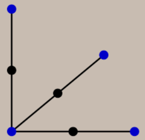

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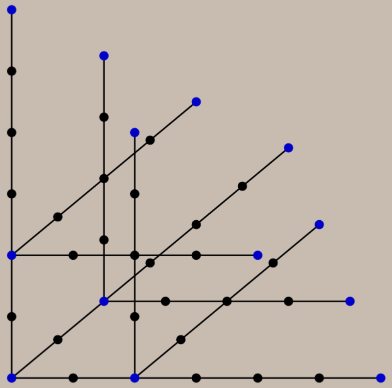

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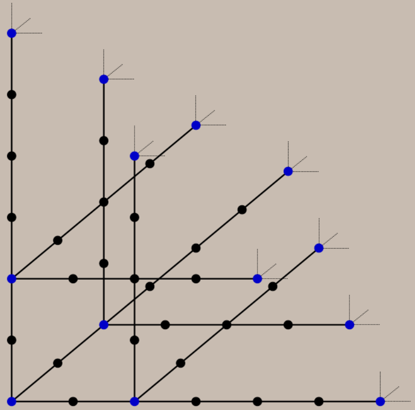

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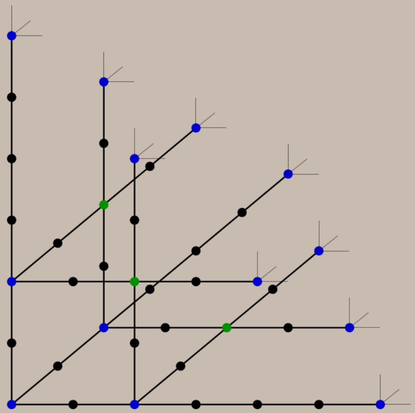

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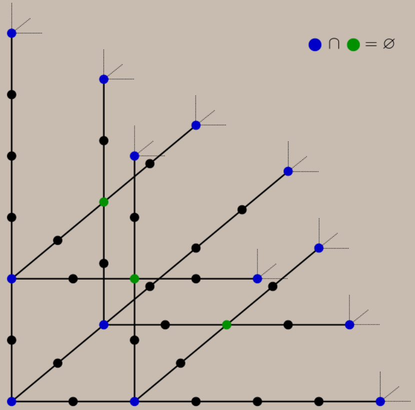

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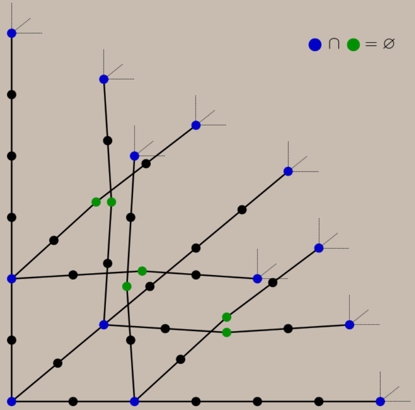

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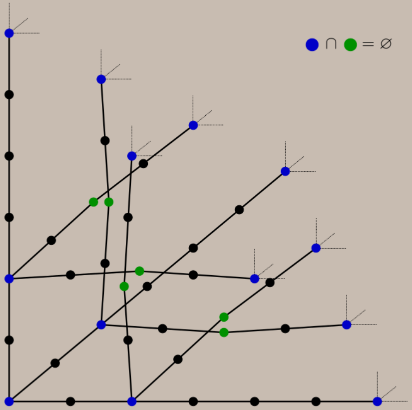

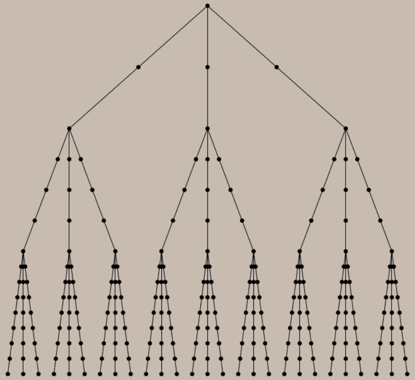

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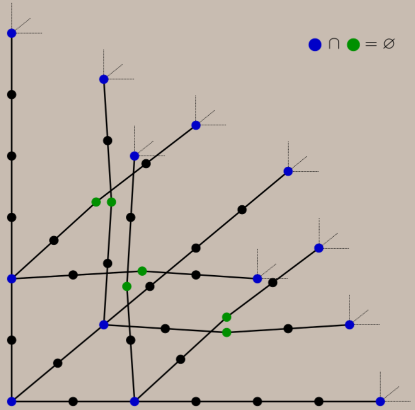
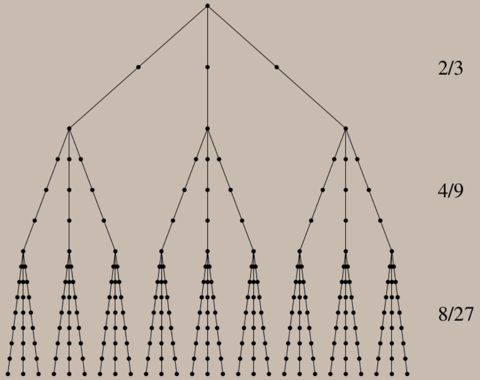
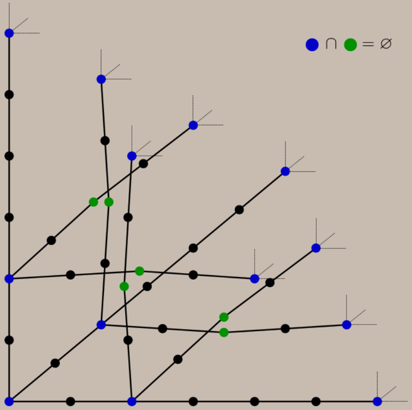


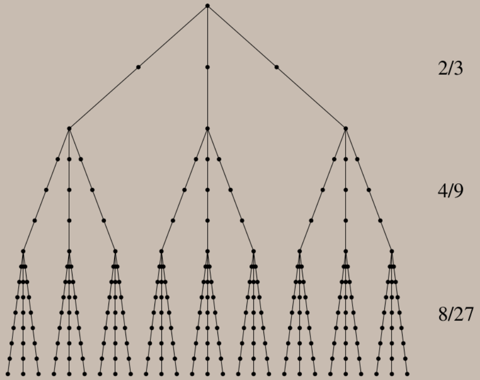


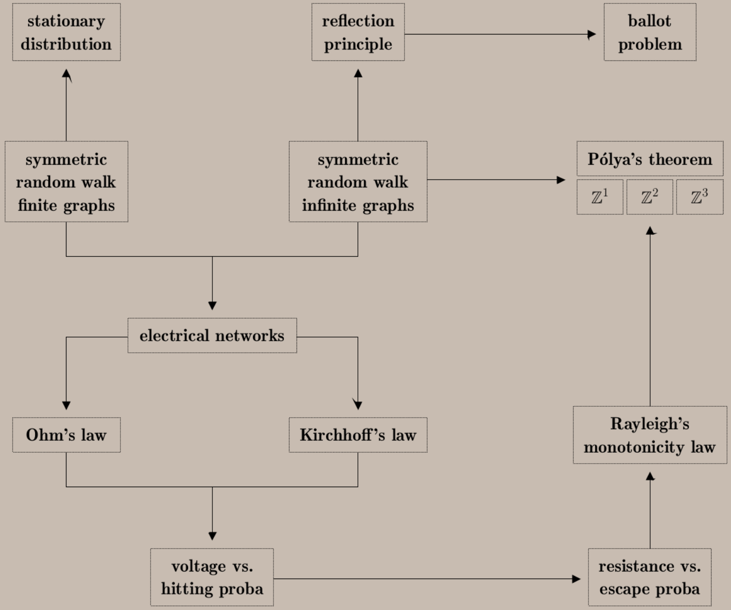
Theorem. $\mathbb{Z}^3$ is transient.

Proof. Cutting = find an infinite subtree embedded in $\mathbb{Z}^3$ with a finite effective resistance.

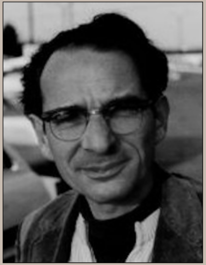
$$R_{eff}(0, r) \leq \frac{2}{3} + \frac{4}{9} + \frac{8}{27} + \cdots + \left(\frac{2}{3}\right)^r \to 2 \quad \text{as the radius } r \to \infty.$$







George Pólya



Frank Spitzer